FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

(PROG 102) PRINCIPLE OF SOFTWARE ENGINEERING

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Class : BSEM1202F

Semester/YEAR : 2/1 **GROUP 8**

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Academic Honesty Policy Statement

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Student's Signature: Date: 08TH MAY 2026

LECTURER'S COMMMENTS/GRADE:	*for office use only upon receive* Remark DATE :………………. TIME : ……………… RECEIVER'S NAME : ……….

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DECLARATION

We hereby declare that this project titled "EduConnect Sierra Leone" is our original work carried out in fulfilment of the requirements for the course PROG 102 – Principles of Software Engineering at Limkokwing University of Creative Technology. The work presented in this report has not been submitted elsewhere for academic purposes.

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We would also like to thank our family, friends, and classmates for their support and motivation during the completion of this project.

Finally, I acknowledge all authors, researchers, and organizations whose publications and resources contributed to the successful completion of this project.

ABSTRACT

Access to educational opportunities, career guidance, scholarships, internships, and mentorship remains a significant challenge for many students and graduates in Sierra Leone. The lack of a centralized digital platform often results in missed opportunities and limited access to information that can support academic and professional growth.

EduConnect Sierra Leone is proposed as a Digital Public Good platform designed to bridge the gap between students, graduates, mentors, employers, and scholarship providers. The platform provides a centralized environment where users can access scholarships, internships, jobs, mentorship programs, and learning resources.

The project follows the Waterfall Software Development Life Cycle (SDLC) model and incorporates various software engineering design techniques including UML diagrams, Data Flow Diagrams (DFDs), Entity Relationship Diagrams (ERDs), System Architecture Design, Network Design, and User Interface Prototyping using Figma.

The proposed solution contributes to Sustainable Development Goal (SDG) 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation and Infrastructure). The platform aims to improve educational accessibility, employability, and career development opportunities for young people across Sierra Leone.

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The rapid advancement of technology has transformed the way people access education, employment opportunities, and professional development resources. Digital platforms have become essential tools for connecting individuals with information and opportunities that can improve their quality of life.

In Sierra Leone, many students and graduates face challenges in accessing scholarships, internships, jobs, mentorship opportunities, and educational resources. Information about these opportunities is often scattered across different websites, social media platforms, institutions, and organizations, making it difficult for young people to find relevant opportunities in a timely manner.

Furthermore, graduates frequently struggle to connect with experienced professionals who can provide career guidance and mentorship. This gap limits career growth, professional networking, and employment opportunities.

To address these challenges, this project proposes EduConnect Sierra Leone, a Digital Public Good platform designed to provide a centralized system where students, graduates, mentors, employers, and scholarship providers can interact and share opportunities. The platform aims to improve access to educational and career development resources while promoting digital inclusion across Sierra Leone.

1.2 Problem Statement

Students and graduates in Sierra Leone face several challenges in accessing educational and career development opportunities. Information about scholarships, internships, jobs, and mentorship programs is often fragmented and difficult to access. As a result, many qualified individuals miss opportunities that could contribute to their academic and professional growth.

Additionally, there is limited interaction between students, graduates, and experienced professionals who can provide mentorship and guidance. The absence of a centralized platform creates inefficiencies in communication, information sharing, and opportunity discovery.

Therefore, there is a need for a unified digital platform that connects students, graduates, mentors, employers, and scholarship providers while simplifying access to educational and career-related opportunities.

1.3 Aim of the Project

The aim of this project is to design a Digital Public Good platform called EduConnect Sierra Leone that connects students, graduates, mentors, employers, scholarship providers, and educational resources through a centralized digital system.

1.4 Objectives of the Project

The objectives of this project are:

1. To provide a centralized platform for educational and career opportunities.

2. To enable students and graduates to discover scholarships and internships.

3. To connect users with experienced mentors for career guidance.

4. To provide access to learning resources and professional development materials.

5. To facilitate communication between users and opportunity providers.

6. To improve employability and educational accessibility through technology.

7. To support sustainable development initiatives through digital innovation.

1.5 Scope of the Project

The scope of EduConnect Sierra Leone covers the design and prototyping of a digital platform that supports students, graduates, mentors, and administrators.

The platform allows users to register and log into the system, search for scholarships, internships, and jobs, access educational resources, connect with mentors, receive notifications, and manage their profiles.

The project focuses on system analysis, system design, architecture design, and prototype development. Actual implementation, deployment, and integration with external organizations are beyond the scope of this project.

1.6 Significance of the Project

This project is significant because it addresses critical challenges faced by students and graduates in Sierra Leone. By providing a centralized platform for opportunities and resources, the system can improve access to education, employment, and mentorship.

The platform also supports digital transformation initiatives by leveraging technology to solve real-world problems. Furthermore, it contributes to national development by promoting skills development, career growth, and lifelong learning opportunities.

The project demonstrates the practical application of software engineering principles in solving social and educational challenges through innovative digital solutions.

CHAPTER 2: REQUIREMENTS ANALYSIS

2.1 Introduction

Requirements analysis is a critical phase of software engineering that focuses on identifying and documenting the needs and expectations of stakeholders. This chapter presents the functional and non-functional requirements of EduConnect Sierra Leone and identifies the primary users of the system.

The analysis provides a foundation for system design and ensures that the proposed solution addresses the challenges identified in the problem statement.

2.2 Stakeholder Analysis

Stakeholder: Students

Role: Access scholarships, internships, jobs, and learning resources

Stakeholder: Graduates

Role: Search for jobs, internships, and mentorship opportunities

Stakeholder: Mentors

Role: Provide career guidance and professional support

Stakeholder: Employers

Role: Advertise job opportunities

Stakeholder: Scholarship Providers

Role: Publish scholarship opportunities

Stakeholder: Administrators

Role: Manage system operations and users.

2.3 Functional Requirements

The functional requirements define the specific services that the system must provide.

1. User Registration

The system shall allow users to create accounts.

2. User Login

The system shall authenticate registered users.

3. Scholarship Search

The system shall allow users to search for scholarships.

4. Job Search

The system shall allow users to search for jobs.

5. Internship Search

The system shall allow users to search for internship opportunities.

6. Mentorship Request

The system shall allow users to request mentorship services.

7. Learning Resources Access

The system shall provide educational materials and resources.

8. Notifications

The system shall send updates and alerts to users.

9. Profile Management

The system shall allow users to update their profiles.

10. Administrative Management

The administrator shall manage users, opportunities, and resources.

2.4 Non-Functional Requirements

1. Security

The system shall protect user information and prevent unauthorized access.

2. Reliability

The system shall operate consistently without failure.

3. Performance

The system shall respond quickly to user requests.

4. Scalability

The system shall support increasing numbers of users.

5. Availability

The system shall be accessible whenever users require services.

6. Usability

The interface shall be simple, intuitive, and user-friendly.

7. Maintainability

The system shall be easy to update and improve.

2.5 System Users

The system consists of four primary user categories:

1. Students

Students access scholarships, internships, learning resources, and mentorship opportunities.

2. Graduates

Graduates search for employment opportunities and professional development resources.

3. Mentors

Mentors provide guidance and support to students and graduates.

4. Administrators

Administrators manage the platform and oversee system activities.

2.6 Feasibility Analysis

Technical Feasibility:

The project is technically feasible because modern web and mobile technologies can support the proposed platform.

Economic Feasibility:

The project requires moderate resources for development and can provide significant educational and social benefits.

Operational Feasibility:

The platform is user-friendly and can be adopted by students, graduates, mentors, and administrators.

Schedule Feasibility:

The project can be completed within the academic timeline provided for the course.

2.7 Requirements Summary

The requirements analysis confirms the need for a centralized digital platform that connects students, graduates, mentors, employers, and scholarship providers. The identified requirements provide a strong foundation for designing a system that improves access to opportunities, educational resources, and professional development services throughout Sierra Leone.

CHAPTER 3: SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

3.1 Introduction

The Software Development Life Cycle (SDLC) is a structured process used in software engineering to plan, design, develop, test, deploy, and maintain software systems. The SDLC provides a systematic approach that helps ensure software projects are completed successfully while meeting user requirements and quality standards.

For the development of EduConnect Sierra Leone, the Waterfall Model was selected because it provides a clear and sequential development process. Each phase must be completed before moving to the next phase, making it suitable for academic projects with clearly defined requirements.

3.2 Waterfall Model

The Waterfall Model is one of the most widely used software development methodologies. It follows a linear sequence of phases where the output of one phase becomes the input for the next phase.

The model was selected for this project because the requirements were clearly identified during the planning stage. It also provides proper documentation and simplifies project management.

Waterfall Model Phases

1. Planning

2. Requirements Gathering

3. System Design

4. Development

5. Testing

6. Deployment

7. Maintenance

3.3 Planning Phase

The planning phase involved identifying the problem, defining project objectives, determining project scope, and establishing the overall project strategy.

During this phase, research was conducted on the challenges faced by students and graduates in Sierra Leone regarding access to scholarships, internships, jobs, mentorship, and educational resources.

The project title, objectives, and expected outcomes were defined to ensure that the proposed solution addressed real-world educational and career development challenges.

3.4 Requirements Gathering Phase

The requirements gathering phase focused on identifying stakeholder needs and system requirements.

The primary stakeholders identified were students, graduates, mentors, scholarship providers, employers, and system administrators.

Functional requirements such as registration, login, scholarship search, job search, mentorship requests, notifications, and profile management were documented. Non-functional requirements including security, reliability, scalability, and usability were also identified.

3.5 System Design Phase

The system design phase involved creating diagrams and models that describe the structure and behavior of the proposed system.

Various software engineering modeling techniques were used including UML diagrams, Data Flow Diagrams, Entity Relationship Diagrams, System Architecture Design, and Network Design.

The design phase ensured that the system requirements were translated into a clear and structured blueprint that could guide future implementation.

3.6 Development Phase

The development phase represents the actual implementation of the software system. Although this project focuses on design and prototyping, the development phase was considered during the architecture and prototype design process.

The proposed system can be implemented using modern web and mobile development technologies such as Flutter, React, Node.js, PHP, MySQL, and cloud-based infrastructure.

The prototype developed using Figma provides a realistic representation of how the final application would function.

3.7 Testing Phase

Testing is essential for ensuring software quality and reliability.

The proposed system would undergo several testing activities including:

1. Unit Testing
2. Integration Testing
3. System Testing
4. User Acceptance Testing

Testing would ensure that all system components function correctly and meet user requirements.

Proposed Test Cases

Test Case	Description	Expected Result	Status
User Registration	User creates an account	Account created successfully	Pass
User Login	User enters valid credentials	Access granted	Pass
Scholarship Search	User searches scholarships	Relevant scholarships displayed	Pass
Internship Search	User searches internships	Internship opportunities displayed	Pass
Job Search	User searches jobs	Job listings displayed	Pass
Mentorship Request	User requests mentorship	Request submitted successfully	Pass
Resource Access	User opens learning resources	Resources displayed	Pass
Notifications	User receives updates	Notifications delivered successfully	Pass
Profile Update	User edits profile information	Changes saved successfully	Pass
Admin Management	Administrator manages users	Records updated correctly	Pass

The proposed test cases provide a structured approach for evaluating system functionality during future implementation and deployment phases.

3.8 Deployment Phase

Deployment involves making the software available to end users.

The proposed EduConnect Sierra Leone platform could be deployed as both a web application and a mobile application. Users would access services through smartphones, tablets, and computers connected to the internet.

Cloud hosting services could be used to improve accessibility and system availability.

3.9 Maintenance Phase

Maintenance involves monitoring and improving the system after deployment.

Future maintenance activities may include:

- Fixing software defects
- Updating content and resources
- Enhancing security
- Adding new features
- Improving system performance

Continuous maintenance ensures that the platform remains useful and relevant to users.

3.10 Summary

The Waterfall Model provided a structured framework for the development of EduConnect Sierra Leone. Each phase contributed to the successful design of the proposed platform and ensured that user requirements were properly addressed.

The completion of the SDLC phases establishes a strong foundation for future implementation and deployment of the system.

CHAPTER 4: SYSTEM DESIGN

4.1 Introduction

System design is the process of defining the architecture, components, interfaces, and data structures of a software system.

This chapter presents the design models developed for EduConnect Sierra Leone. The diagrams provide a visual representation of system functionality, behavior, data flow, architecture, and infrastructure.

These models serve as blueprints for the proposed solution and demonstrate the application of software engineering principles.

4.2 **Use Case Diagram**

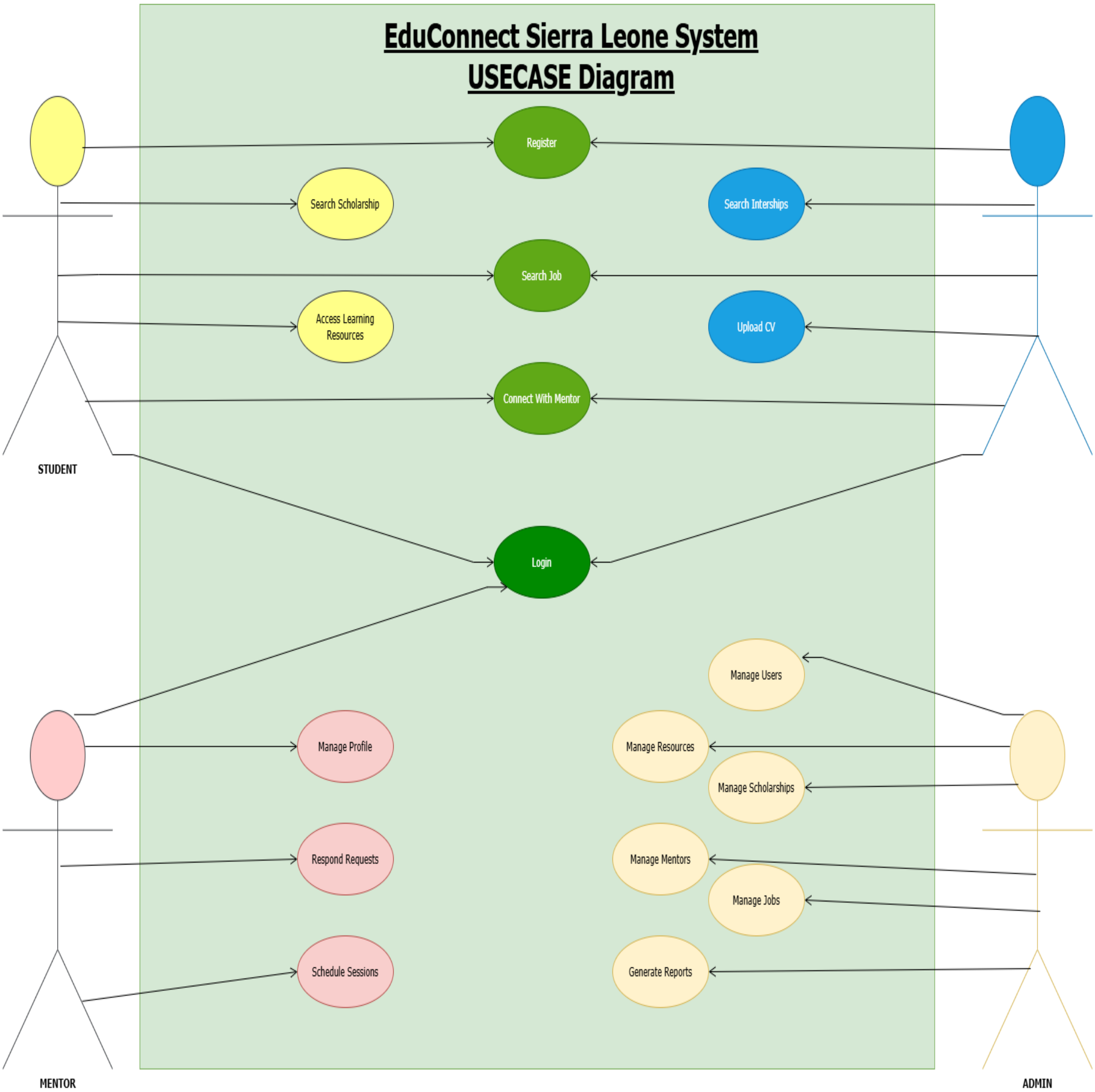

Figure 4.1: Use Case Diagram of EduConnect Sierra Leone

The Use Case Diagram illustrates the interactions between users and the system. The primary actors include Students, Graduates, Mentors, and Administrators.

The diagram shows how users interact with system functionalities such as registration, login, scholarship search, job search, internship search, mentorship requests, learning resource access, profile management, and administrative functions.

4.3 **Activity Diagram**

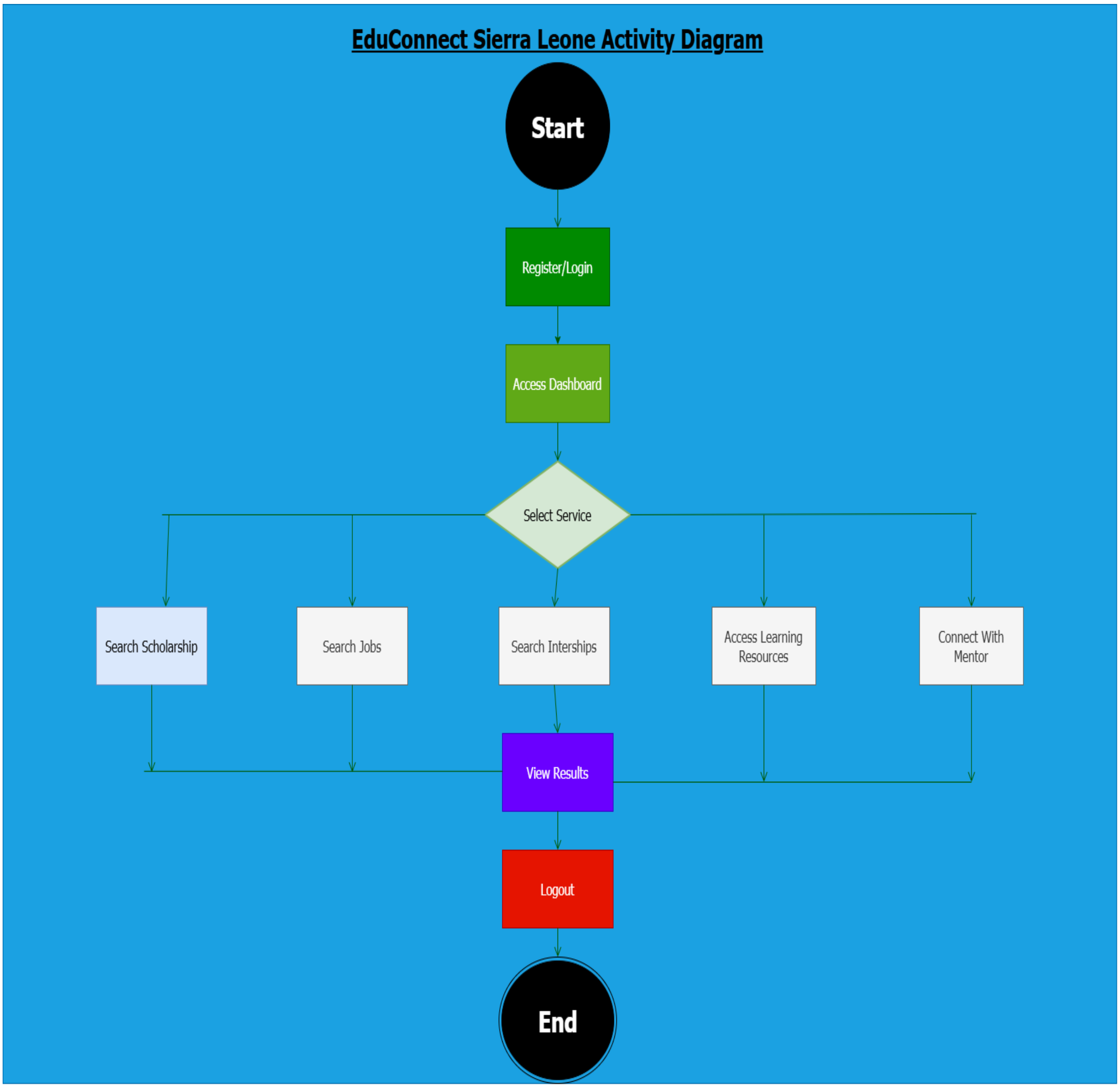

Figure 4.2: Activity Diagram of EduConnect Sierra Leone

The Activity Diagram illustrates the workflow of user activities within the system. It demonstrates the sequence of actions performed by users from registration and authentication to accessing opportunities and resources.

The diagram provides a clear representation of process flow and decision points within the system.

4.4 **Sequence Diagram**

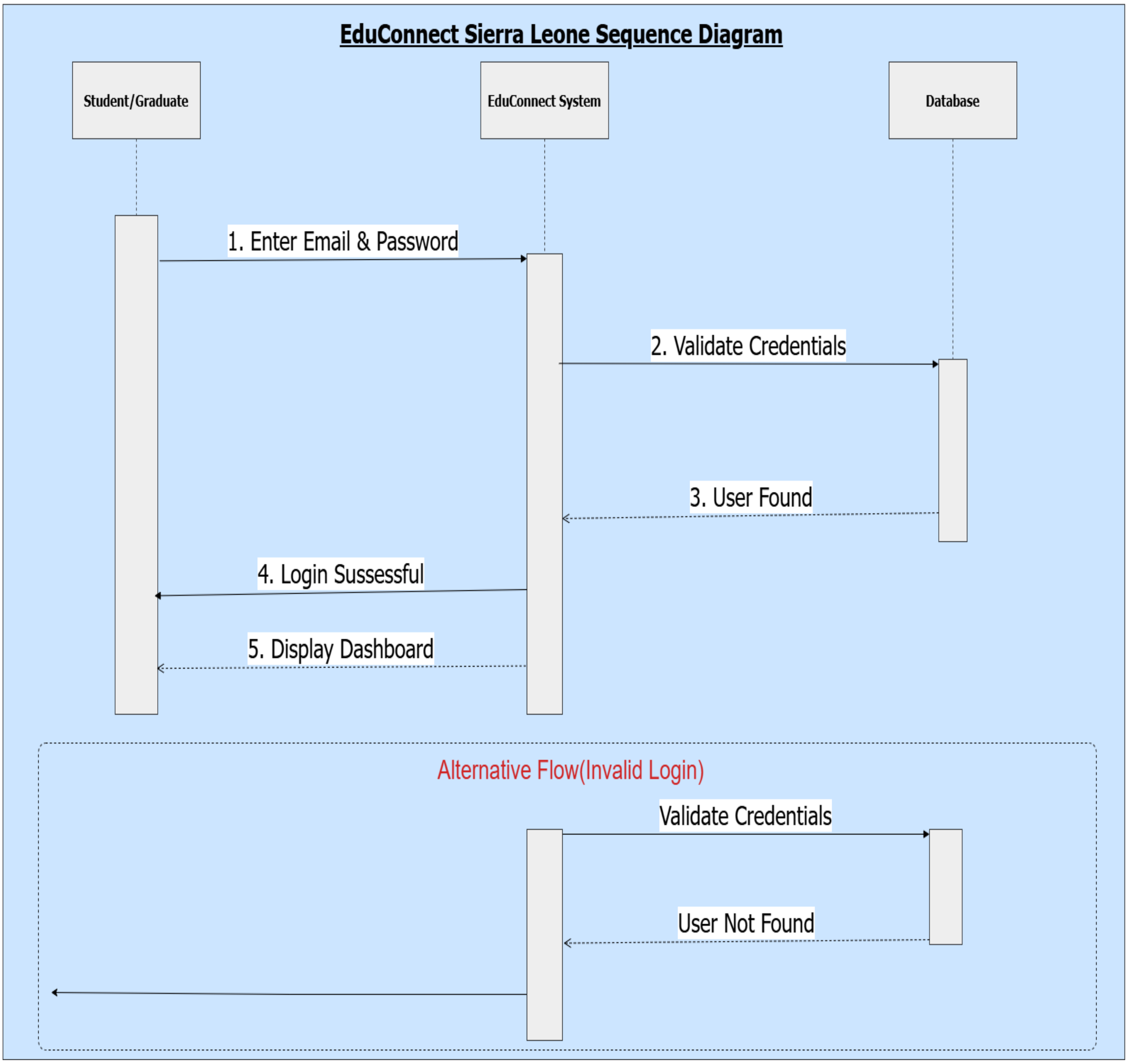

Figure 4.3: Sequence Diagram of EduConnect Sierra Leone

The Sequence Diagram shows how objects interact over time during a typical user session.

It illustrates message exchanges between users, system components, and the database while performing activities such as login, searching for opportunities, and accessing resources.

4.5 **Class Diagram**

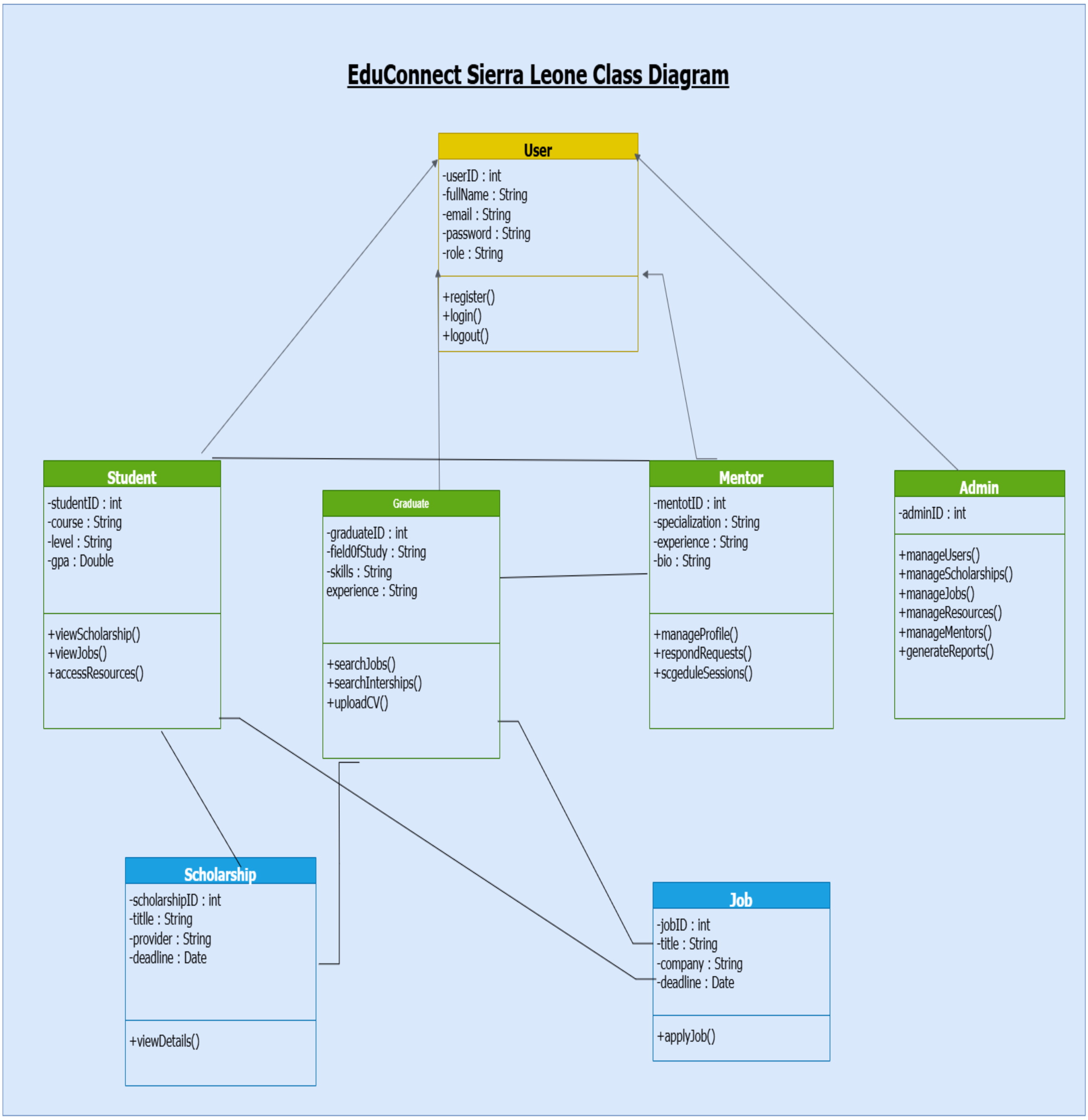

Figure 4.4: Class Diagram of EduConnect Sierra Leone

The Class Diagram represents the static structure of the system.

The diagram includes major classes such as User, Student, Graduate, Mentor, Administrator, Scholarship, and Job. It also illustrates attributes, methods, inheritance relationships, and associations between classes.

4.6 **State Diagram**

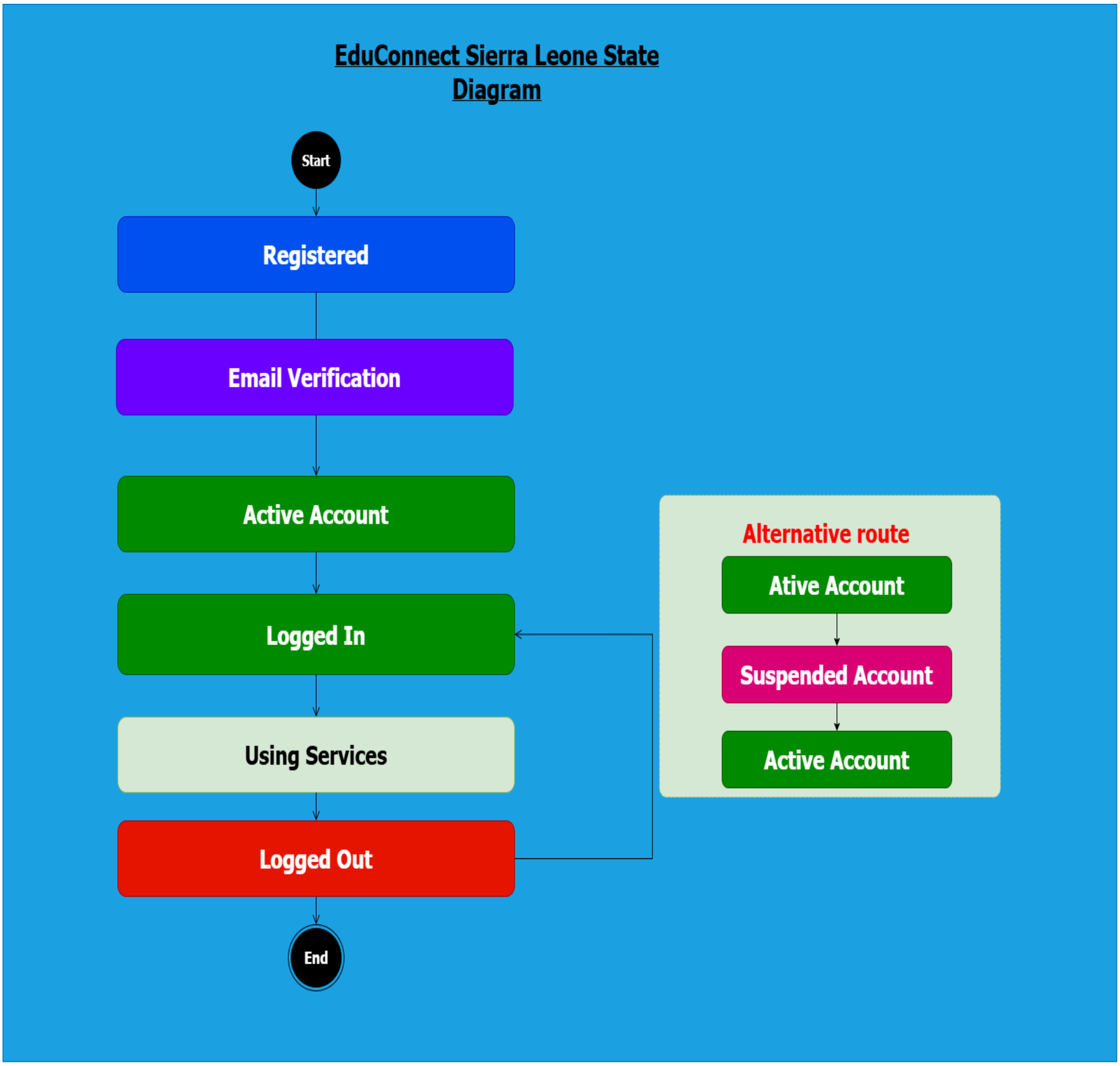

Figure 4.5: State Diagram of EduConnect Sierra Leone

The State Diagram illustrates the different states of a user account throughout its lifecycle.

States include registration, verification, activation, login, service usage, logout, and account suspension. The diagram helps describe how the system responds to user actions and events.

4.7 **Entity Relationship Diagram (ERD)**

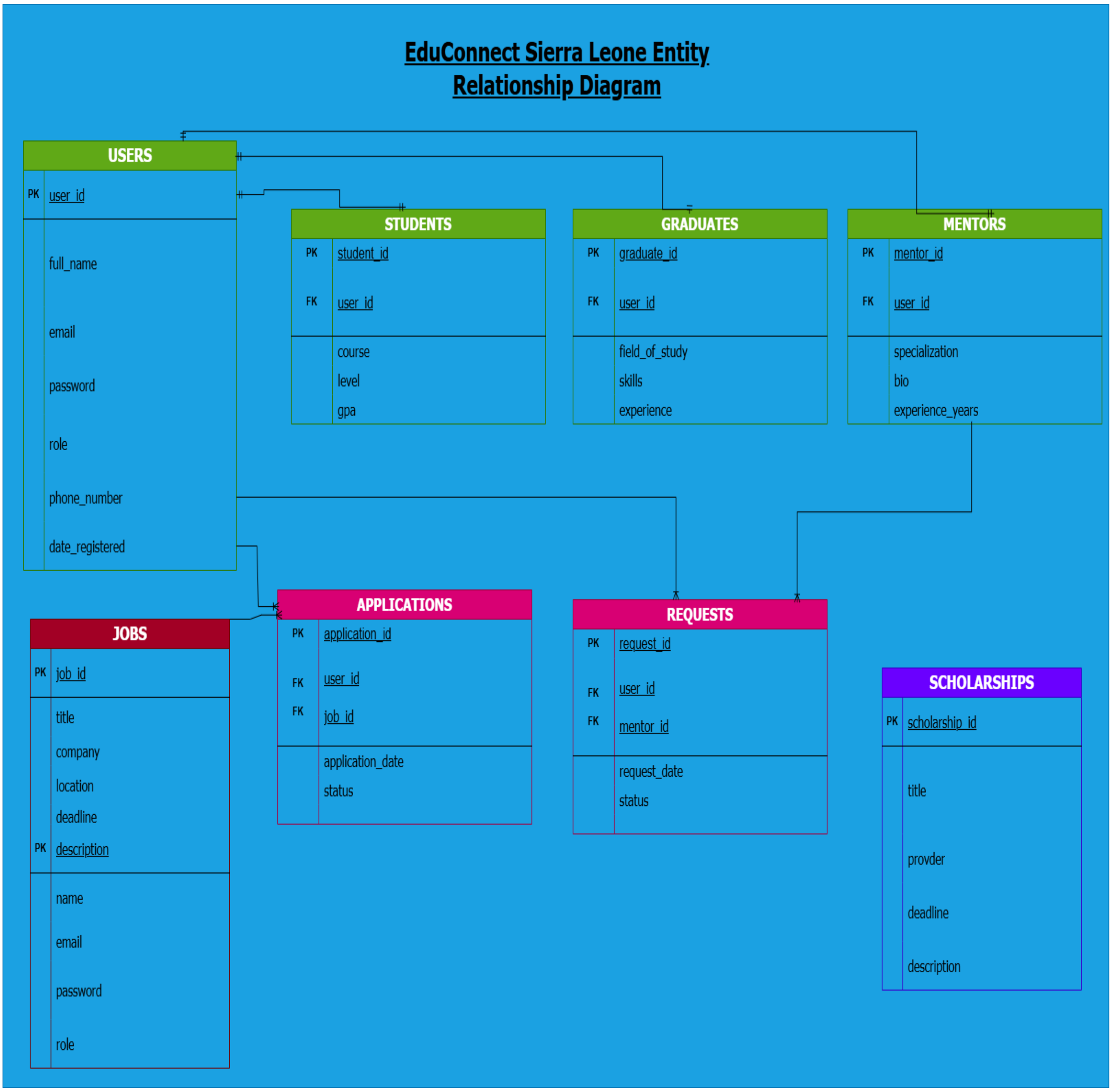

Figure 4.6: Entity Relationship Diagram of EduConnect Sierra Leone

The Entity Relationship Diagram defines the database structure of the system.

The ERD identifies entities such as Users, Students, Graduates, Mentors, Scholarships, Jobs, Applications, and Mentor Requests. It also shows primary keys, foreign keys, and relationships among entities.

4.8 **Data Flow Diagram (DFD) Level 0**

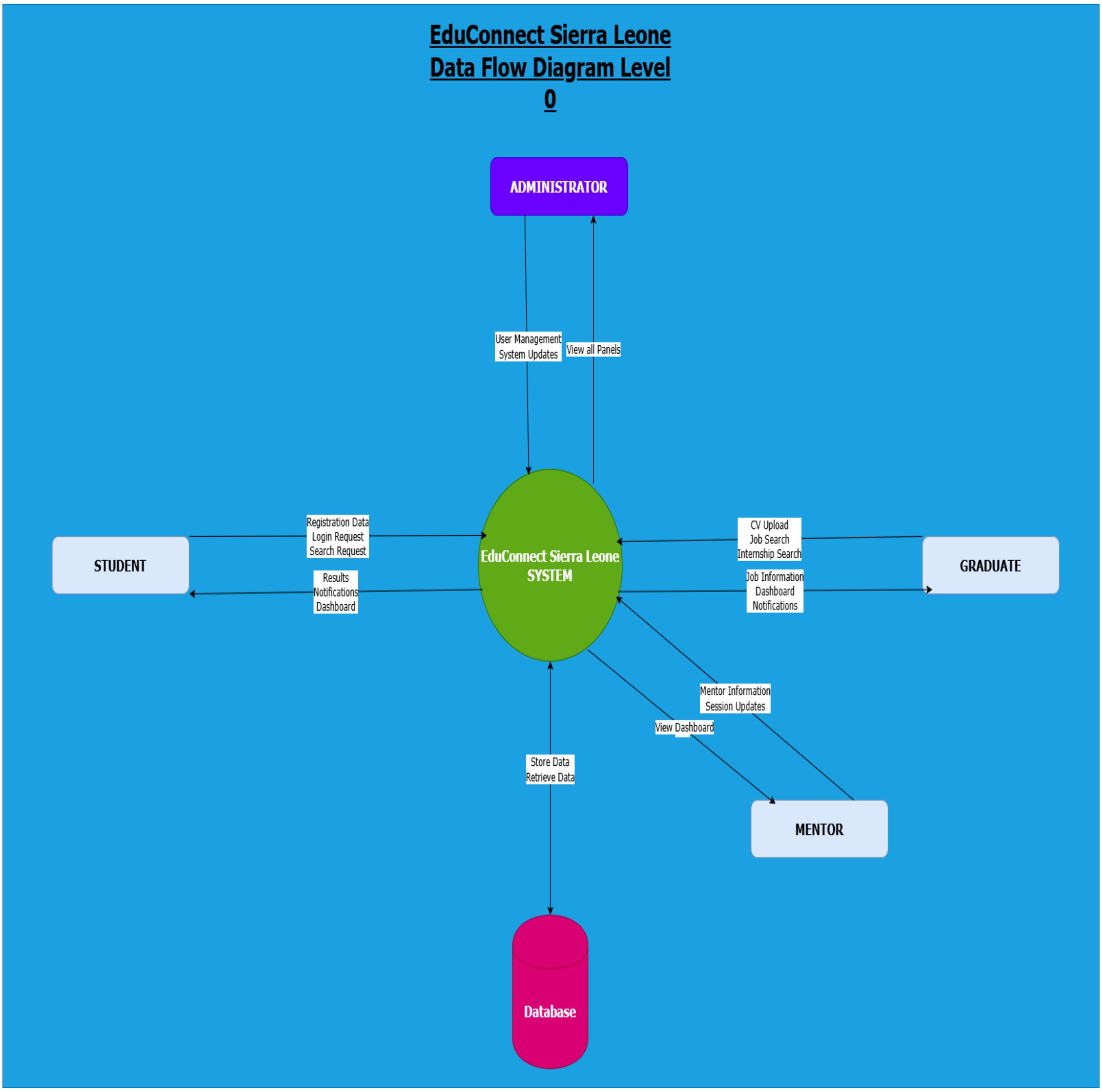

Figure 4.7: Data Flow Diagram Level 0

The DFD Level 0, also known as the Context Diagram, presents the system as a single process and illustrates the interaction between external entities and the system.

The diagram identifies the flow of information between users and the EduConnect Sierra Leone platform.

4.9 **Data Flow Diagram (DFD) Level 1**

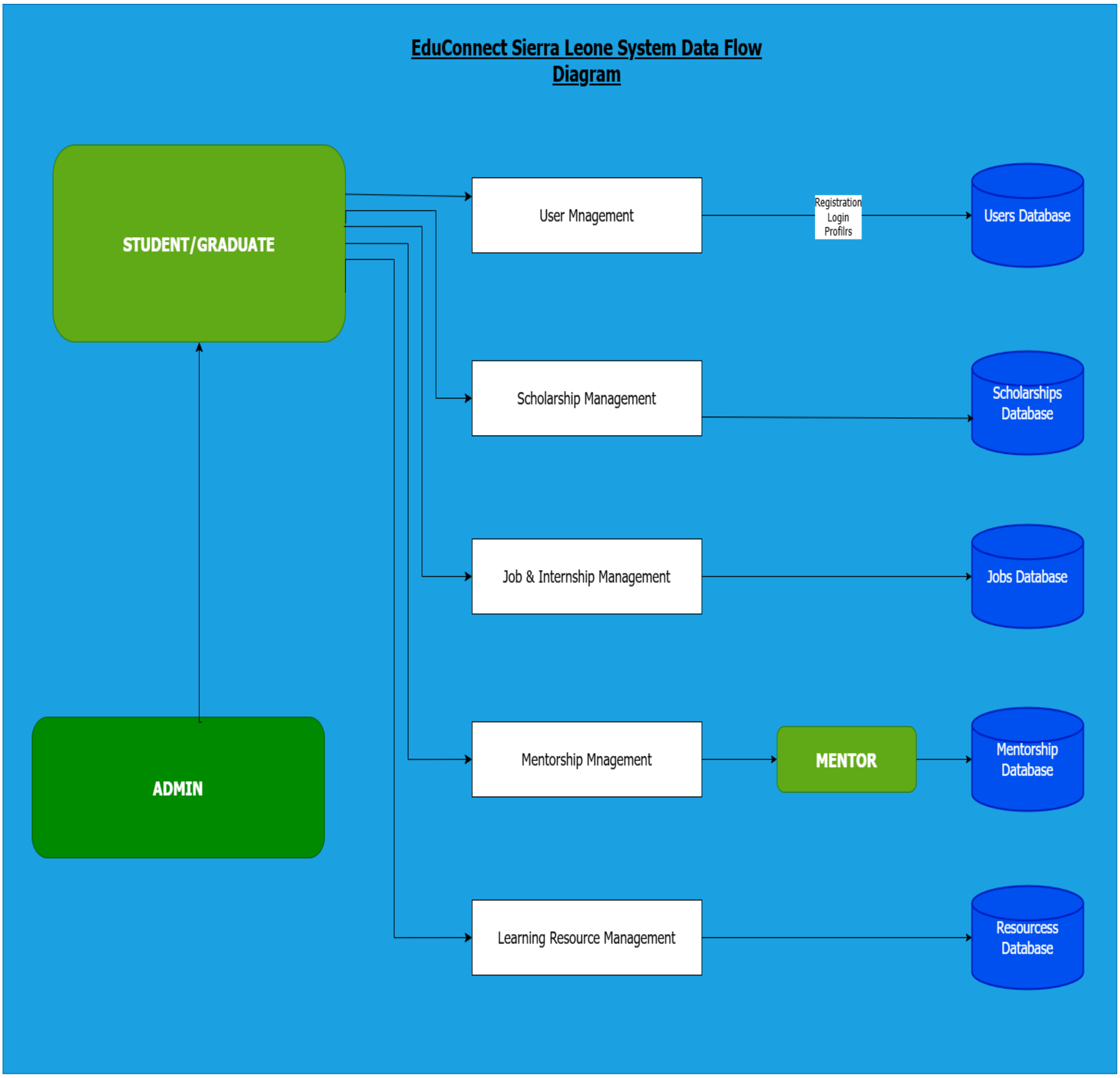

Figure 4.8: Data Flow Diagram Level 1

The DFD Level 1 expands the context diagram by decomposing the main system into several processes.

The diagram illustrates processes such as User Management, Scholarship Management, Job Management, Mentorship Management, and Learning Resource Management.

4.10 **System Architecture Diagram**

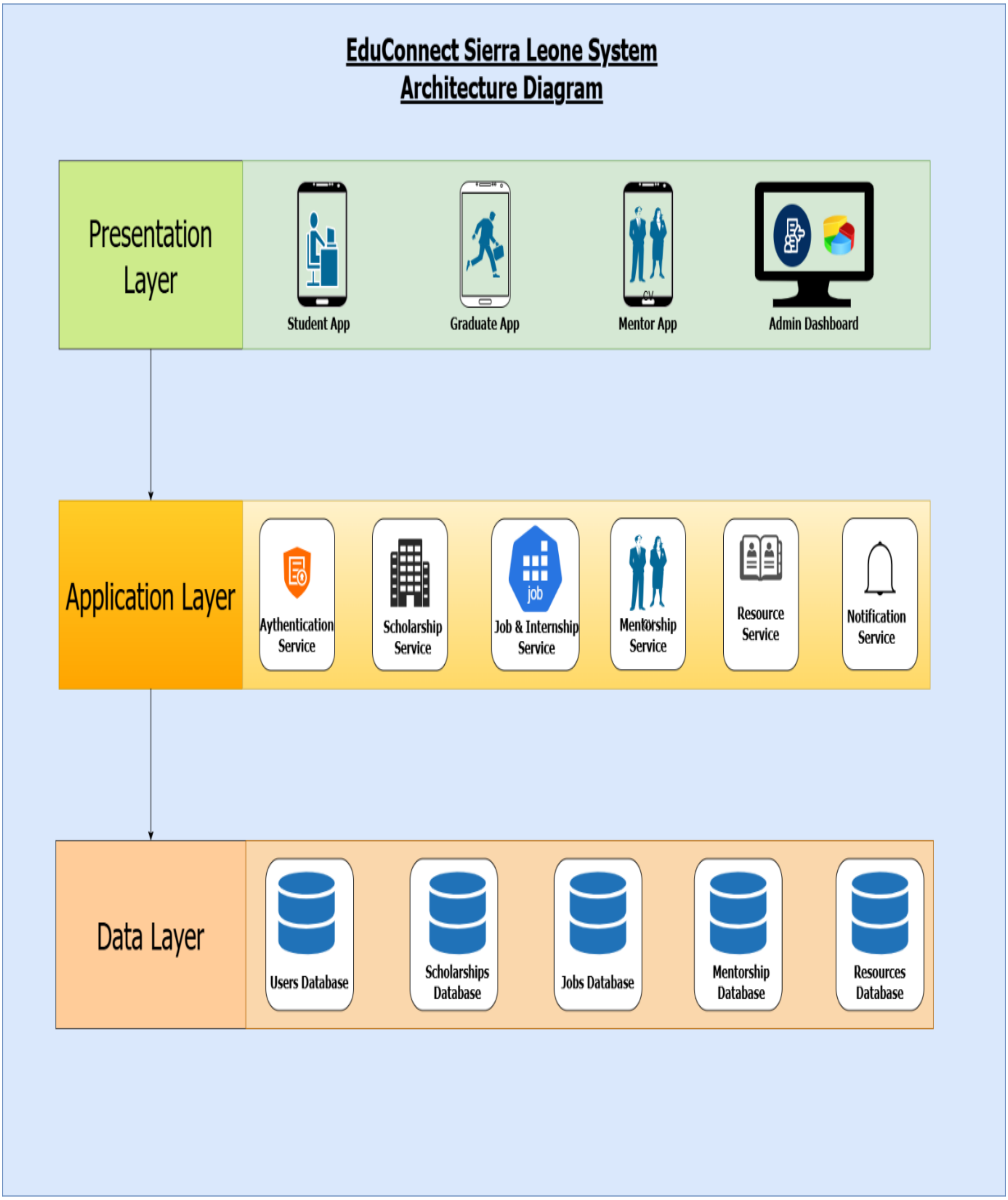

Figure 4.9: System Architecture Diagram

The System Architecture Diagram presents the overall structure of the proposed solution.

4.10.1 Component Interaction

The EduConnect Sierra Leone platform follows a layered architecture where different system components interact to process user requests and provide services efficiently.

The interaction begins when a user accesses the platform through a mobile application or web browser. The user's request is first received by the Presentation Layer, which serves as the interface between users and the system.

When a user attempts to log in, the Presentation Layer sends the login credentials to the Authentication Service located in the Application Layer. The Authentication Service validates the information by communicating with the Database in the Data Layer. If the credentials are valid, the system grants access and redirects the user to the Dashboard.

When users search for scholarships, internships, or job opportunities, the request is forwarded to the appropriate service module. For example, scholarship-related requests are handled by the Scholarship Service, while employment opportunities are managed by the Job Service. These services retrieve relevant records from the database and return the results to the Presentation Layer for display.

Similarly, mentorship requests are processed through the Mentorship Service. Users can browse mentor profiles, submit mentorship requests, and receive responses through the platform. The Mentorship Service stores and retrieves mentorship information from the database.

The Notification Service continuously monitors system events and generates alerts related to application deadlines, scholarship updates, mentorship activities, and system announcements. Notifications are then delivered to users through the Presentation Layer.

Administrators interact with the system through the Admin Dashboard, where they can manage users, opportunities, educational resources, and system reports. Administrative actions are processed through the Application Layer and reflected in the Data Layer.

This interaction model ensures efficient communication between system components while improving scalability, maintainability, and overall system performance.

Component Interaction Flow

1. User accesses the application.
2. Request is received by the Presentation Layer.
3. Presentation Layer forwards request to the Application Layer.
4. Appropriate service processes the request.
5. Service communicates with the Data Layer.
6. Database returns requested information.
7. Service processes response.

8. Presentation Layer displays results to the user.

The architecture consists of three layers:

1. **Presentation Layer:** The Presentation Layer serves as the interface through which users interact with the system. Students, graduates, mentors, and administrators access the application through mobile devices or web browsers.

When a user performs an action such as logging in, searching for scholarships, applying for internships, or requesting mentorship, the request is sent to the Application Layer

2. **Application Layer:** The Application Layer processes the request through the appropriate service module, such as Authentication Service, Scholarship Service, Job Service, Mentorship Service, or Notification Service.

These services communicate with the Data Layer to retrieve, update, or store information in the database.

3. **Data Layer:** The Data Layer stores all user information, scholarship records, job listings, mentorship requests, learning resources, and application data.

The processed information is then returned to the Presentation Layer and displayed to the user.

This layered architecture improves maintainability, scalability, and system organization.

4.11 **Network Diagram**

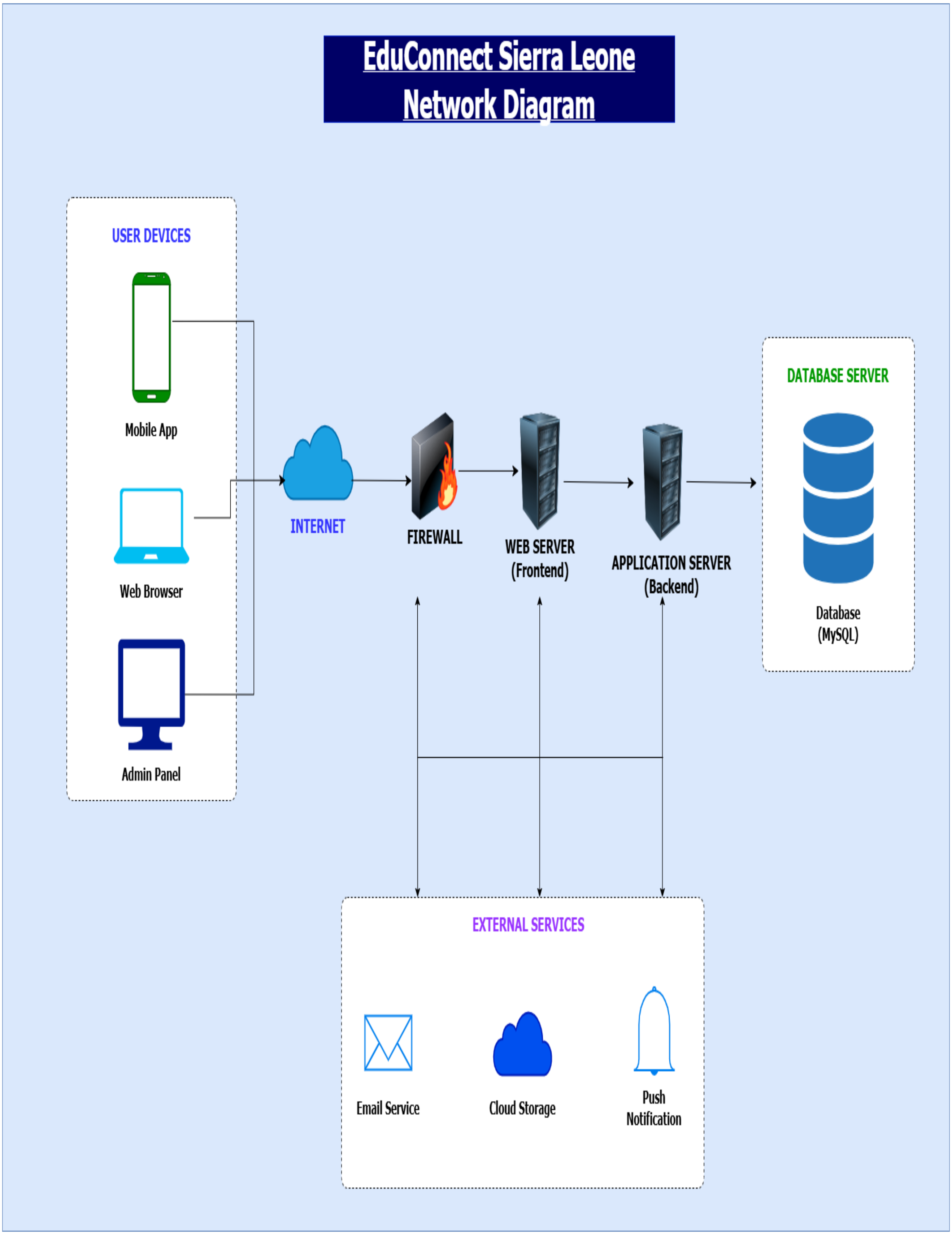

Figure 4.10: Network Diagram

The Network Diagram illustrates how users connect to the system through the internet.

The diagram includes user devices, internet connectivity, firewall protection, web servers, application servers, database servers, and external services.

This design ensures secure communication and reliable access to system resources.

4.12 Chapter Summary

This chapter presented the complete system design for EduConnect Sierra Leone.

Various software engineering models and diagrams were developed to describe system functionality, behavior, architecture, and infrastructure. These design artifacts provide a comprehensive blueprint for the future implementation of the platform.

CHAPTER 5: PROTOTYPE DEVELOPMENT

5.1 Introduction

Prototype development is an important phase in software engineering that allows designers and stakeholders to visualize how a system will look and function before implementation.

For this project, a high-fidelity mobile application prototype was designed using Figma. The prototype demonstrates the user interface, navigation flow, and functionality of EduConnect Sierra Leone.

The prototype was developed to provide users with an intuitive and user-friendly experience while ensuring accessibility, consistency, and ease of navigation.

5.2 Prototype Design Tools

The prototype was developed using Figma, a cloud-based user interface and user experience design tool.

Figma was selected because it provides:

- Collaborative design capabilities
- Interactive prototyping features
- Responsive mobile screen design
- Modern UI/UX components
- Easy export and sharing options

The prototype follows modern mobile application design principles and uses a consistent color scheme, typography, icons, and navigation structure.

5.3 Design Principles Applied

The following design principles were applied during prototype development:

1. **Simplicity**

The interface was designed to be clean and easy to understand.

2. **Consistency**

Colors, typography, icons, and layouts remain consistent throughout the application.

3. **Accessibility**

The design ensures that information is easy to read and interact with.

4. **User-Centered Design**

The prototype focuses on the needs of students, graduates, mentors, and administrators.

5. **Responsive Layout**

The design follows mobile-first principles and is suitable for smartphones.

5.4 **Splash Screen**

Figure 5.1: Splash Screen

The Splash Screen is the first screen displayed when the application launches.

It introduces the EduConnect Sierra Leone brand and presents the application logo and slogan, "Learn • Connect • Grow." This screen helps establish the application's identity and provides a professional first impression.

5.5 **Welcome Screen**

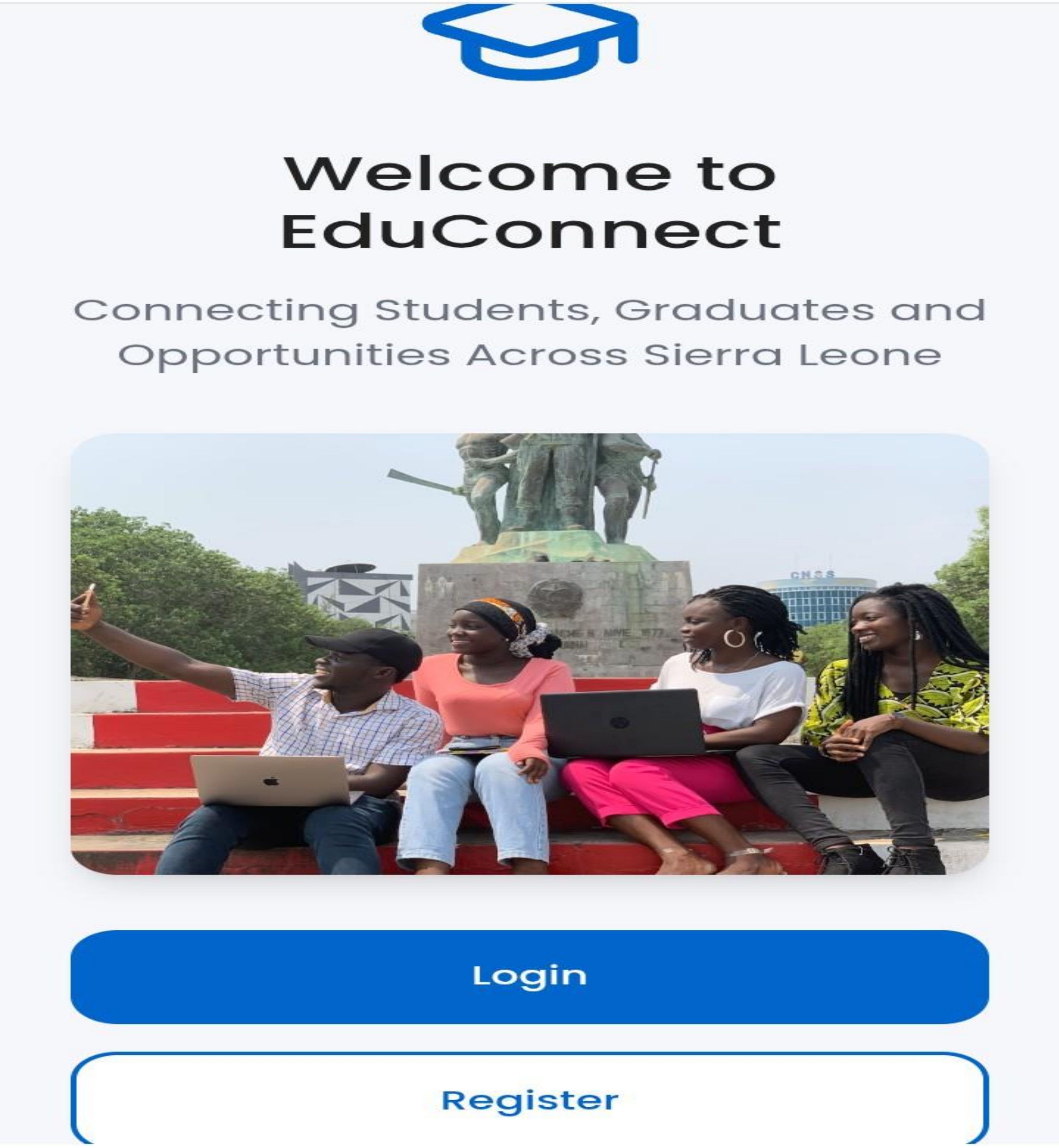

Figure 5.2: Welcome Screen

The Welcome Screen introduces users to the purpose of the application.

It contains a brief description of the platform and provides navigation options for users to either log in or create a new account.

5.6 **Login Screen**

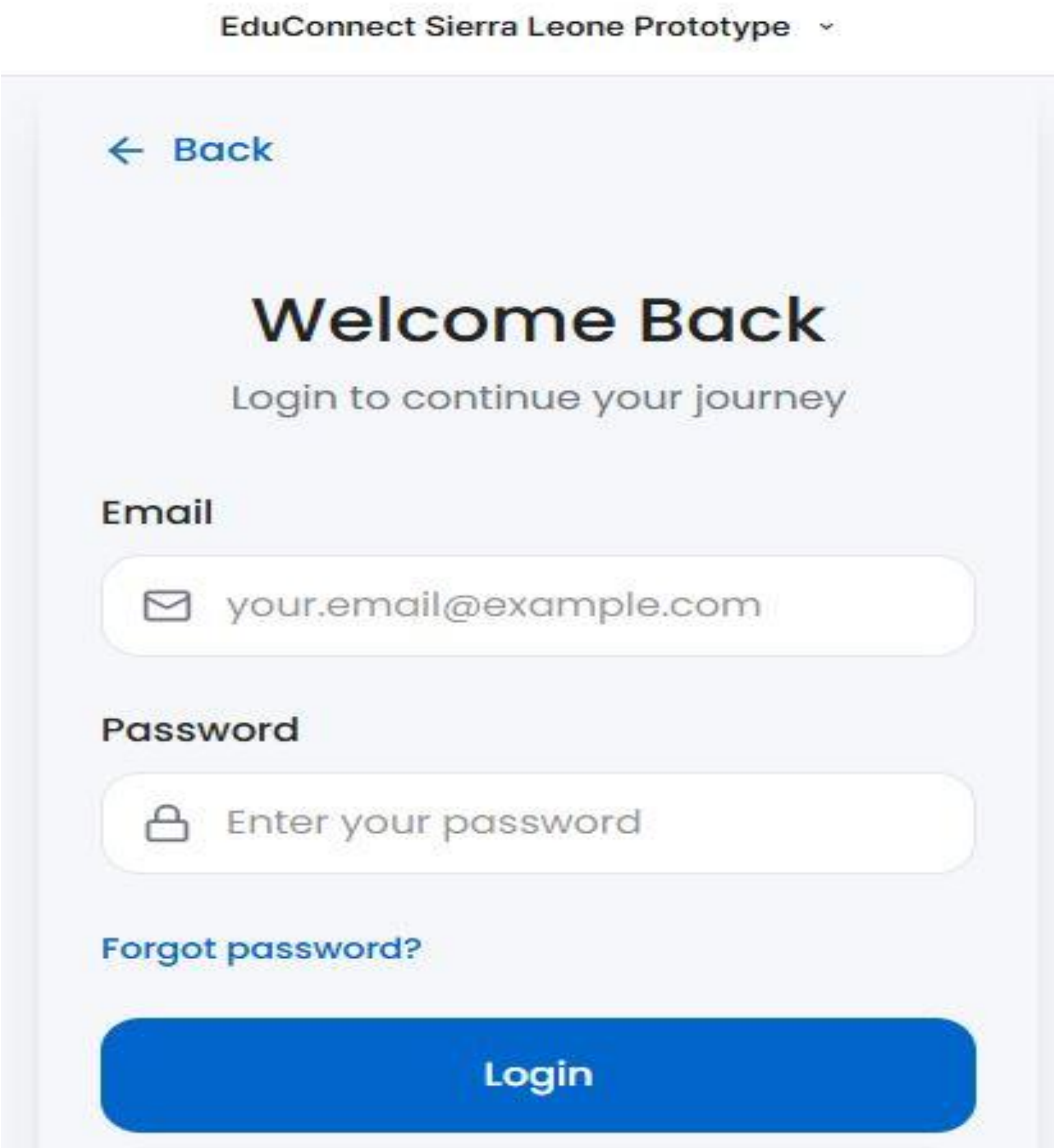

Figure 5.3: Login Screen

The Login Screen allows registered users to securely access the system.

Users enter their email address and password to authenticate their identity and gain access to the platform's services.

5.7 **Registration Screen**

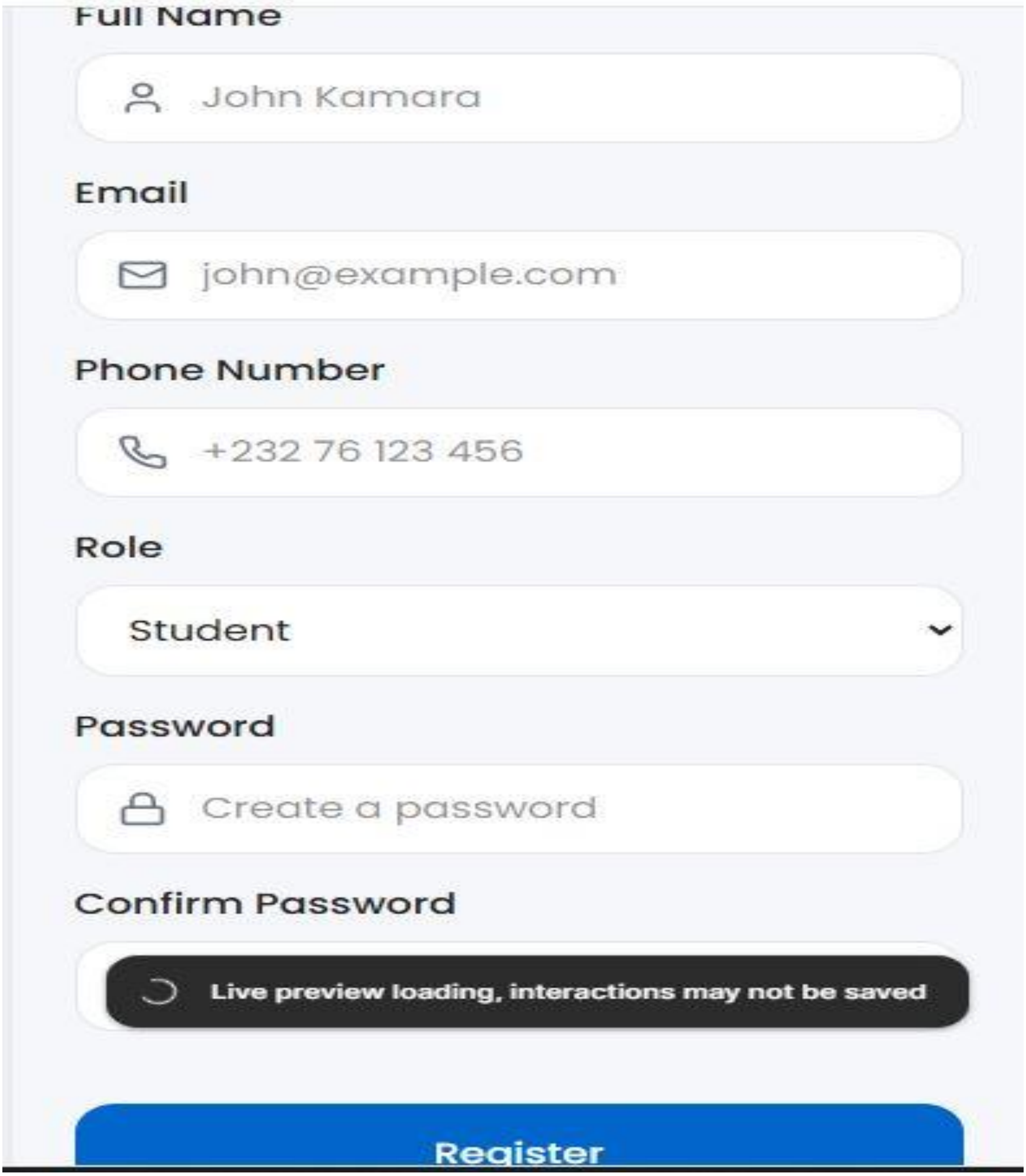

Figure 5.4: Registration Screen

The Registration Screen enables new users to create accounts.

Users provide personal information including full name, email address, phone number, role selection, and password information.

5.8 **Dashboard Screen**

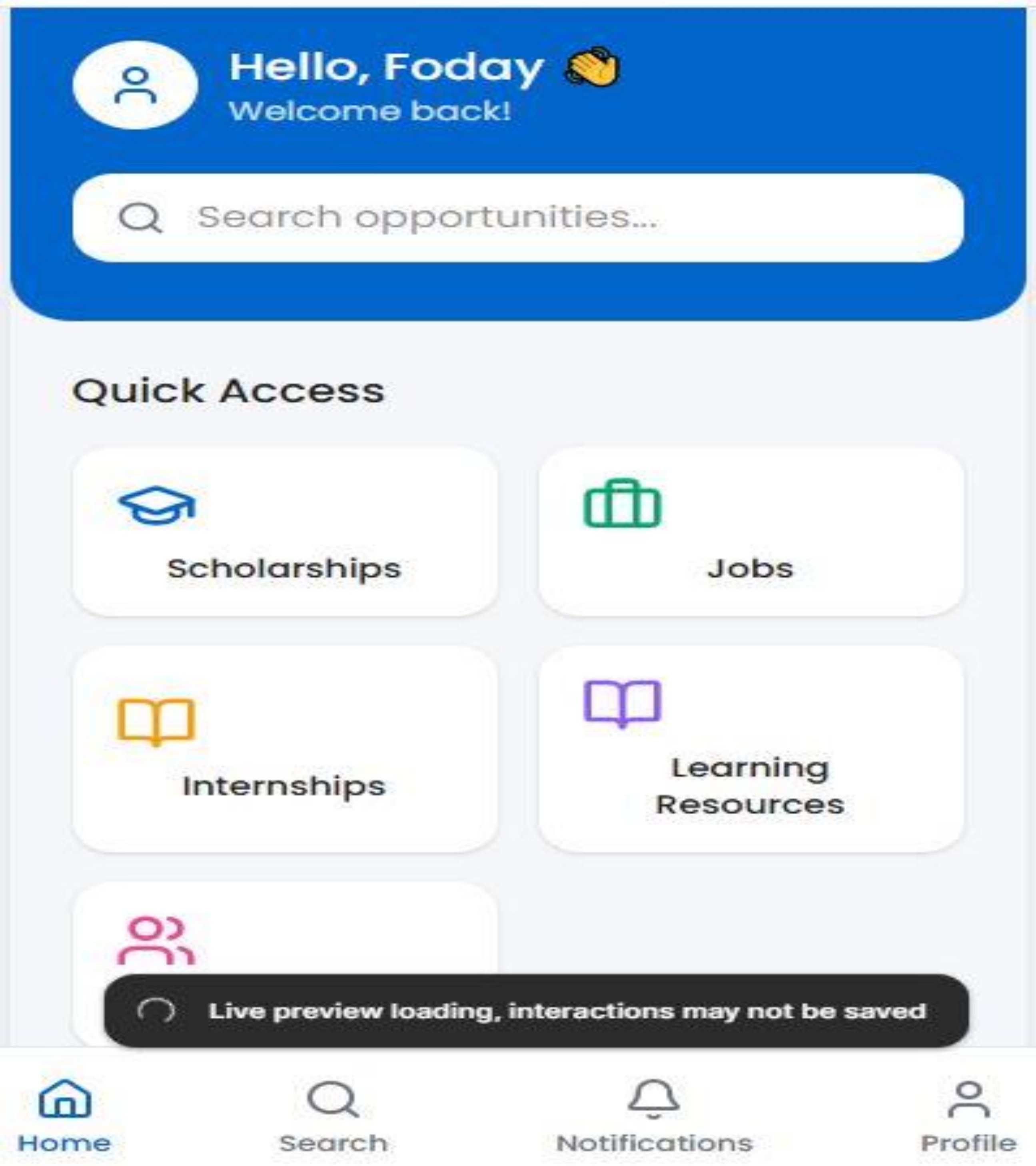

Figure 5.5: Dashboard Screen

The Dashboard serves as the central hub of the application.

It provides quick access to scholarships, jobs, internships, mentors, learning resources, notifications, and profile management functions.

5.9 **Scholarships Screen**

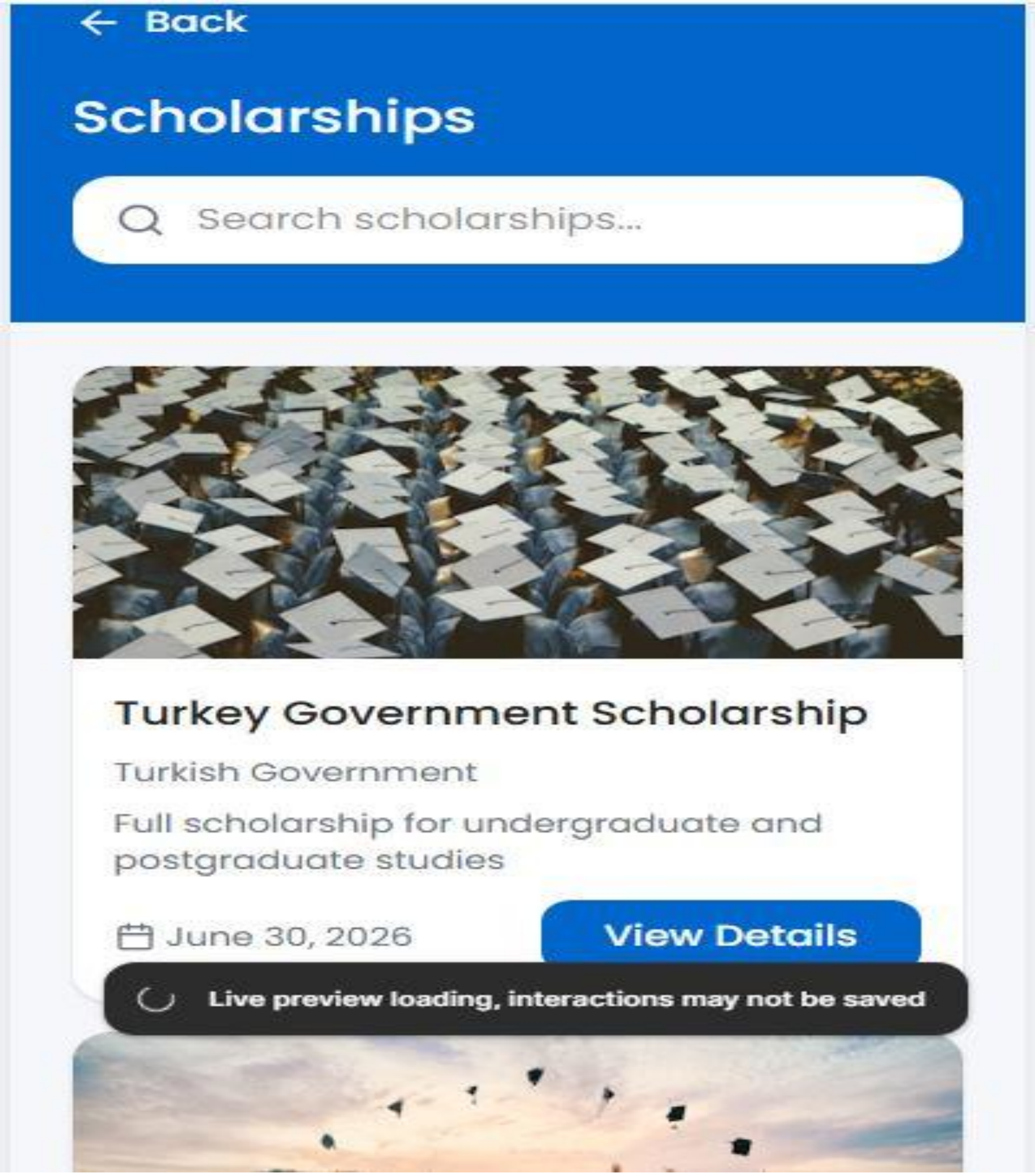

Figure 5.6: Scholarships Screen

The Scholarships Screen provides users with access to scholarship opportunities.

Users can search for scholarships, view details, review eligibility requirements, and identify application deadlines.

5.10 **Jobs Screen**

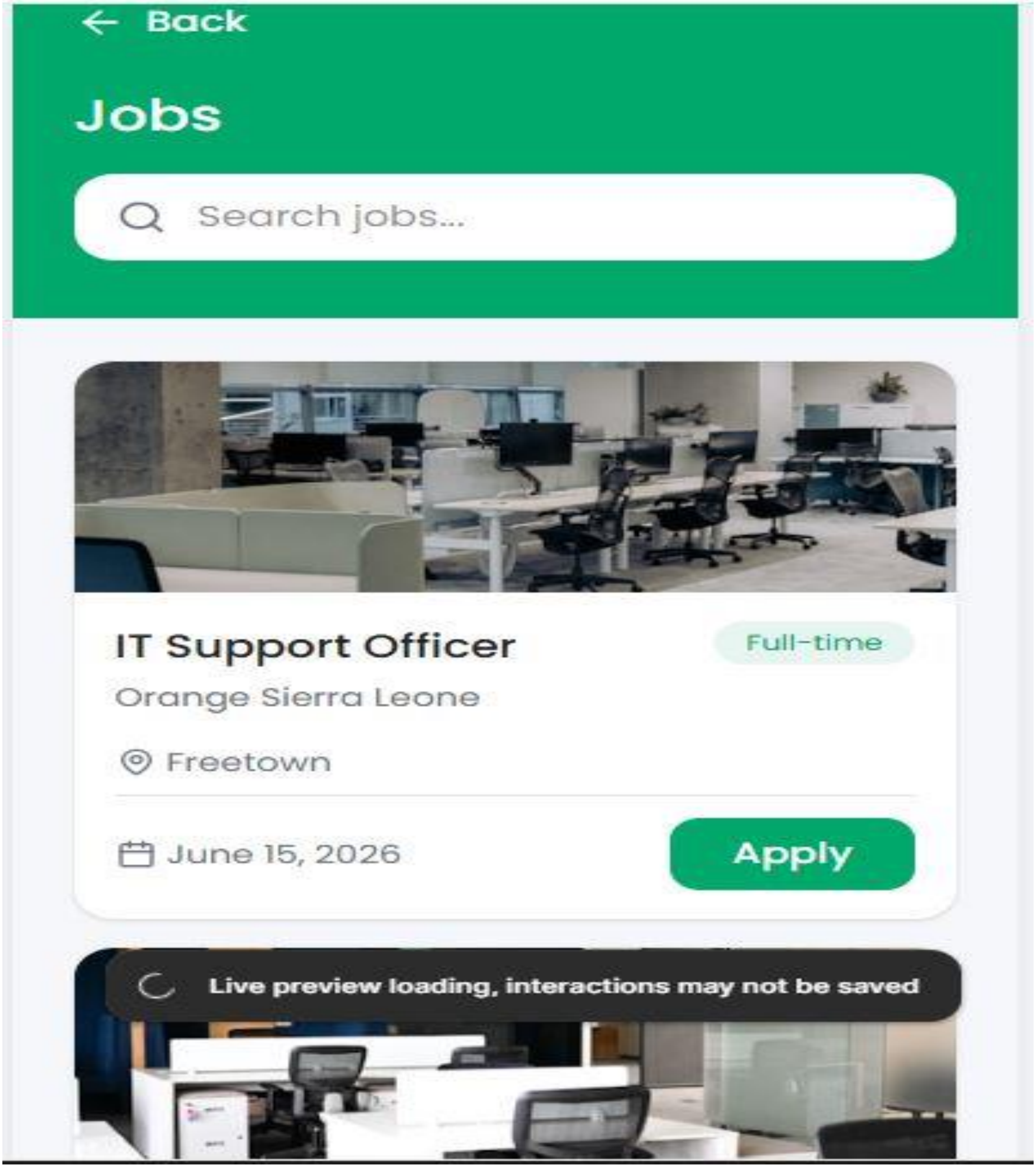

Figure 5.7: Jobs Screen

The Jobs Screen allows users to explore employment opportunities.

Job listings display relevant information such as employer details, deadlines, and application options.

5.11 **Internships Screen**

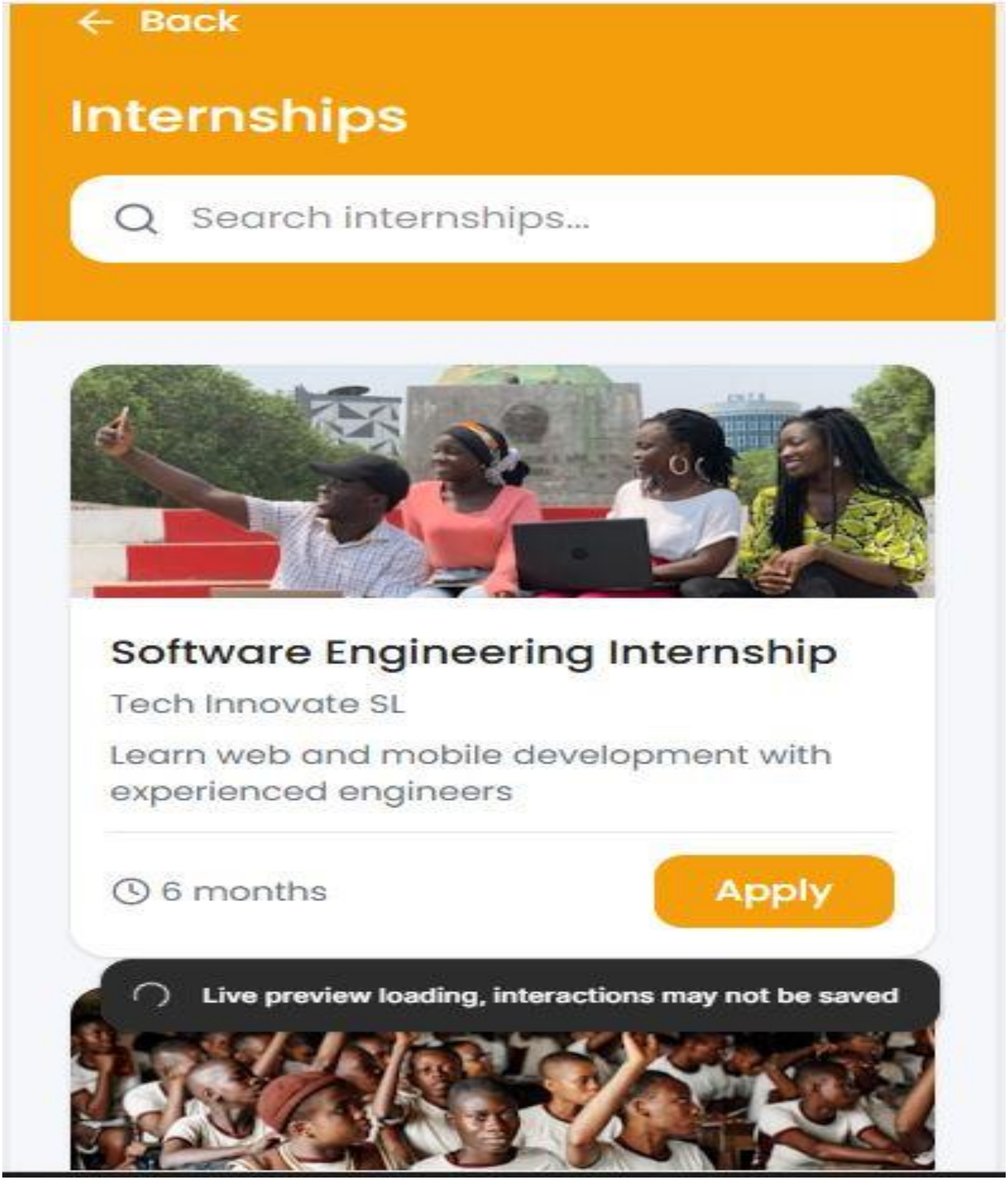

Figure 5.8: Internships Screen

The Internships Screen provides internship opportunities that help students and graduates gain practical experience and improve employability.

5.12 **Learning Resources Screen**

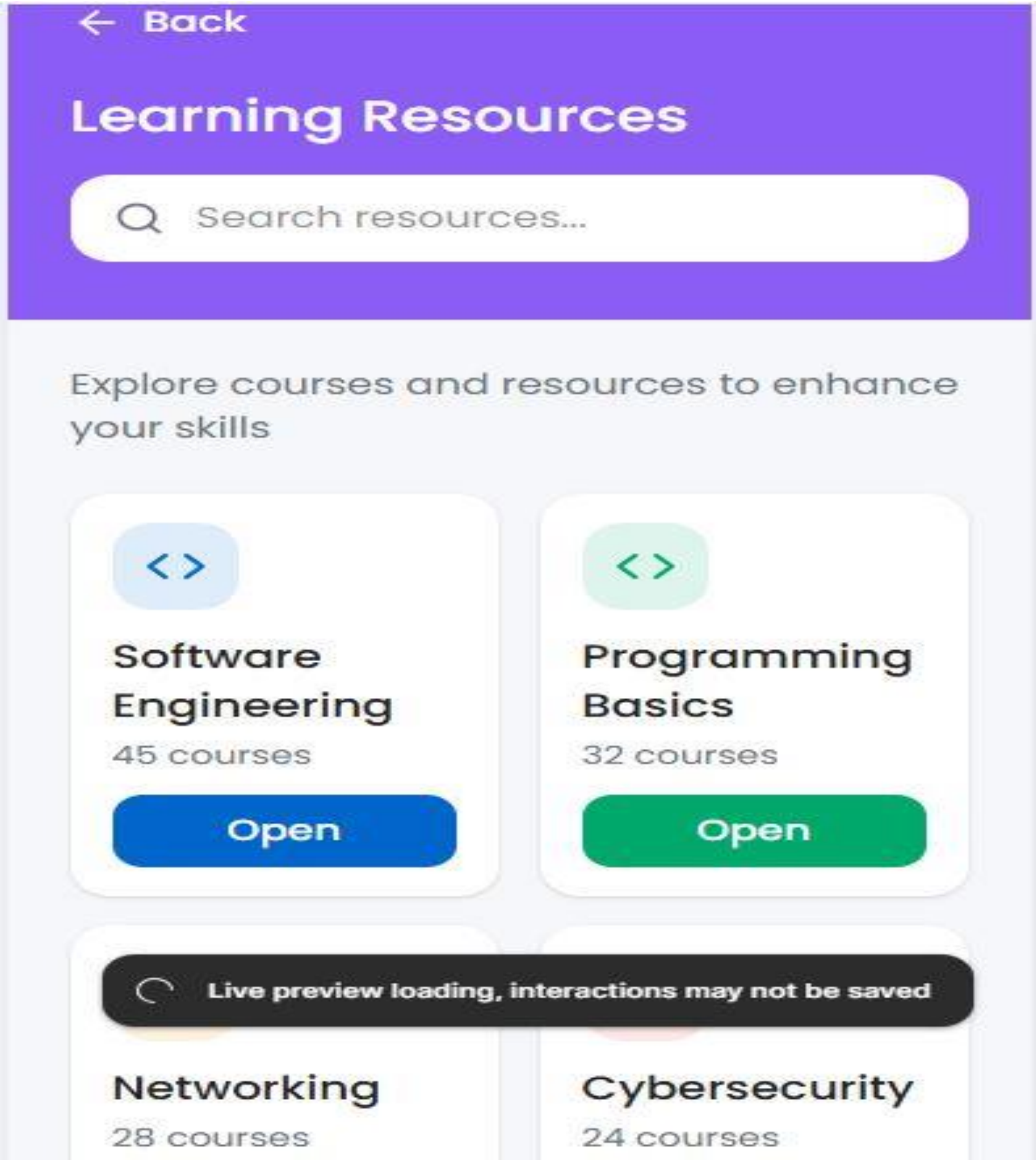

Figure 5.9: Learning Resources Screen

The Learning Resources Screen provides educational materials, online courses, tutorials, and professional development content.

The resources support continuous learning and skill development.

5.13 **Mentor Directory Screen**

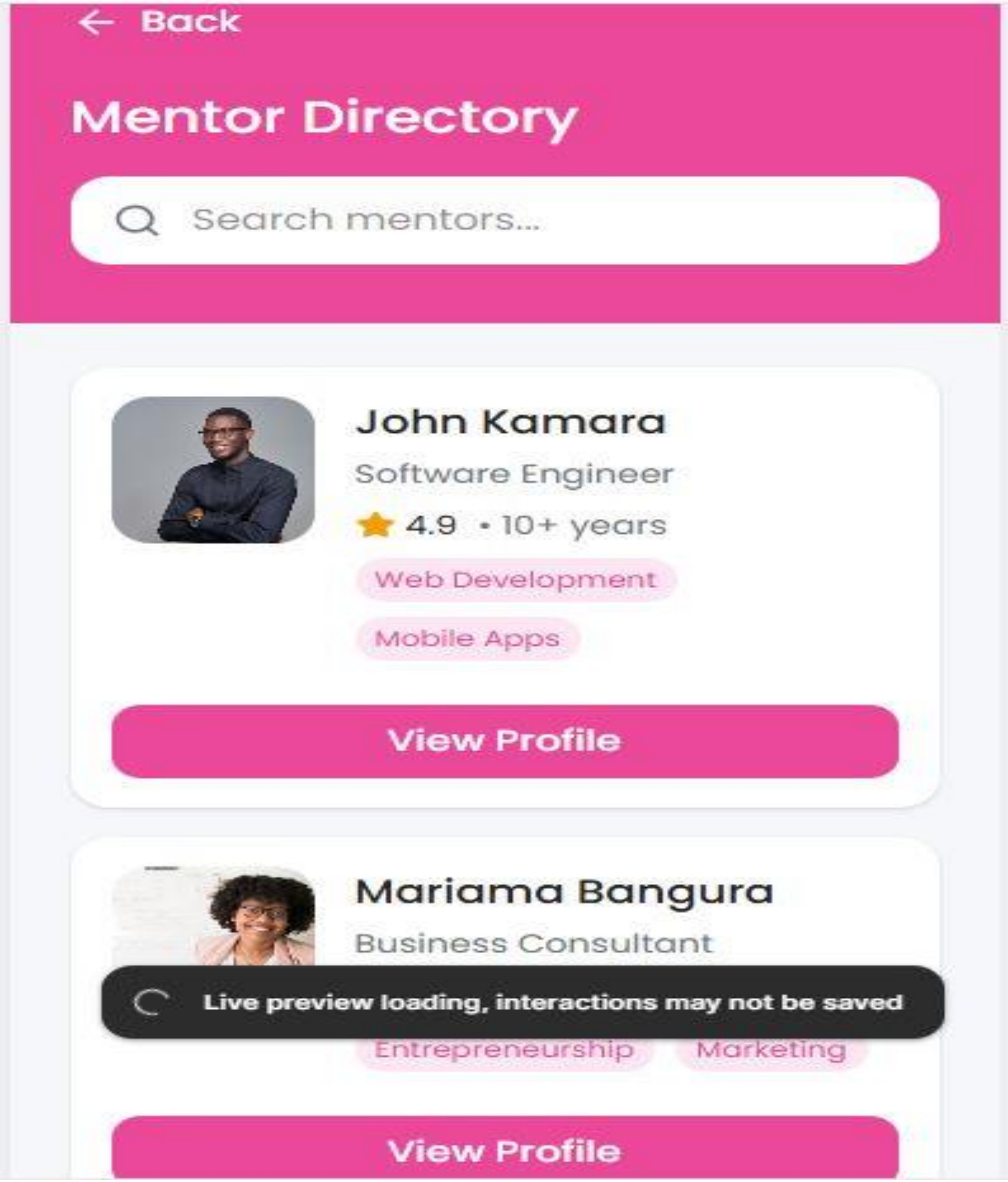

Figure 5.10: Mentor Directory Screen

The Mentor Directory allows users to browse available mentors based on expertise, specialization, experience, and ratings.

5.14 **Mentor Profile Screen**

Figure 5.11: Mentor Profile Screen

The Mentor Profile Screen provides detailed information about a selected mentor.

Users can review qualifications, professional experience, specializations, and mentorship availability.

5.15 **My Applications Screen**

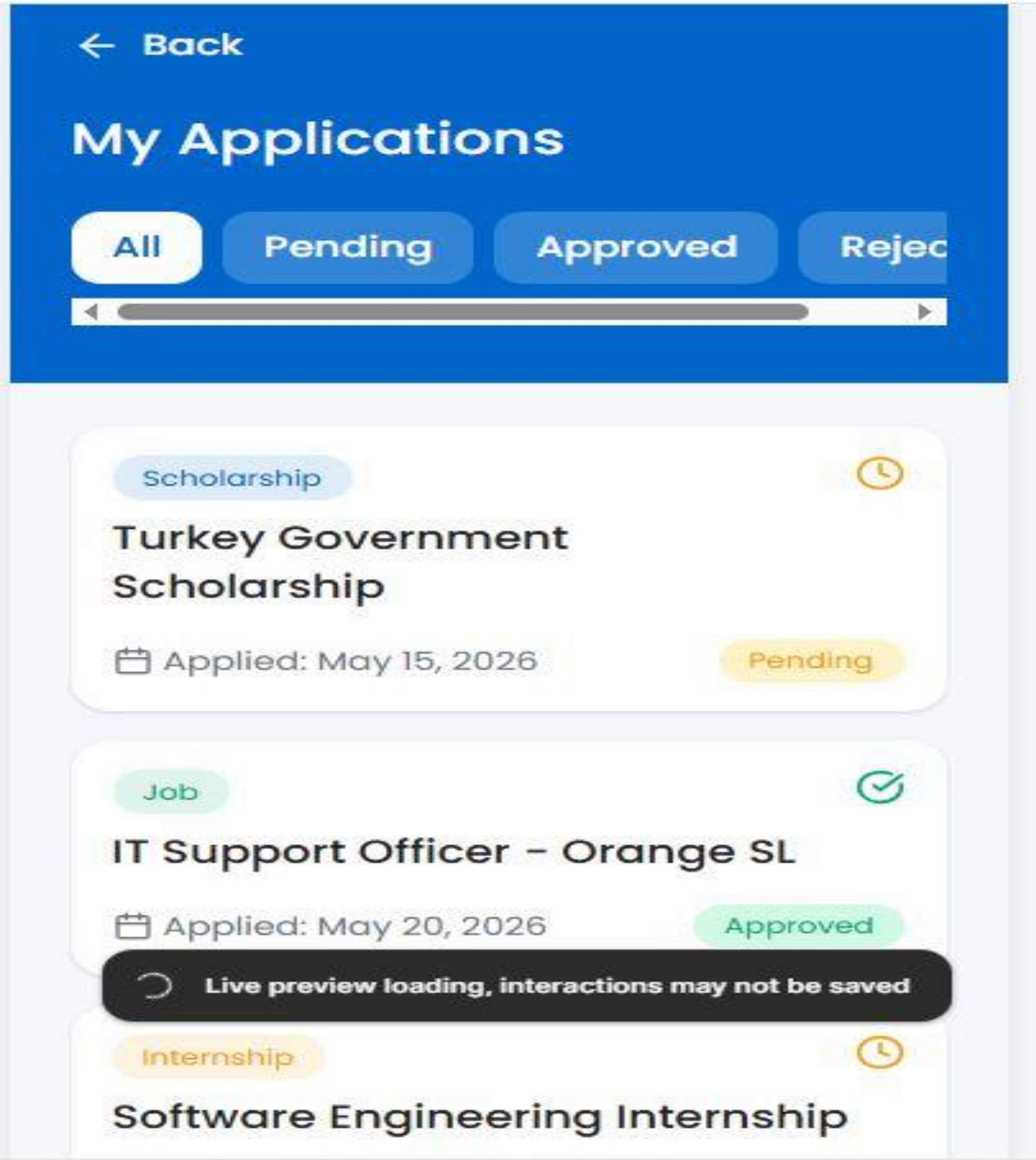

Figure 5.12: My Applications Screen

The My Applications Screen enables users to monitor applications submitted for scholarships, internships, and job opportunities.

Application statuses are displayed to help users track progress.

5.16 **Notifications Screen**

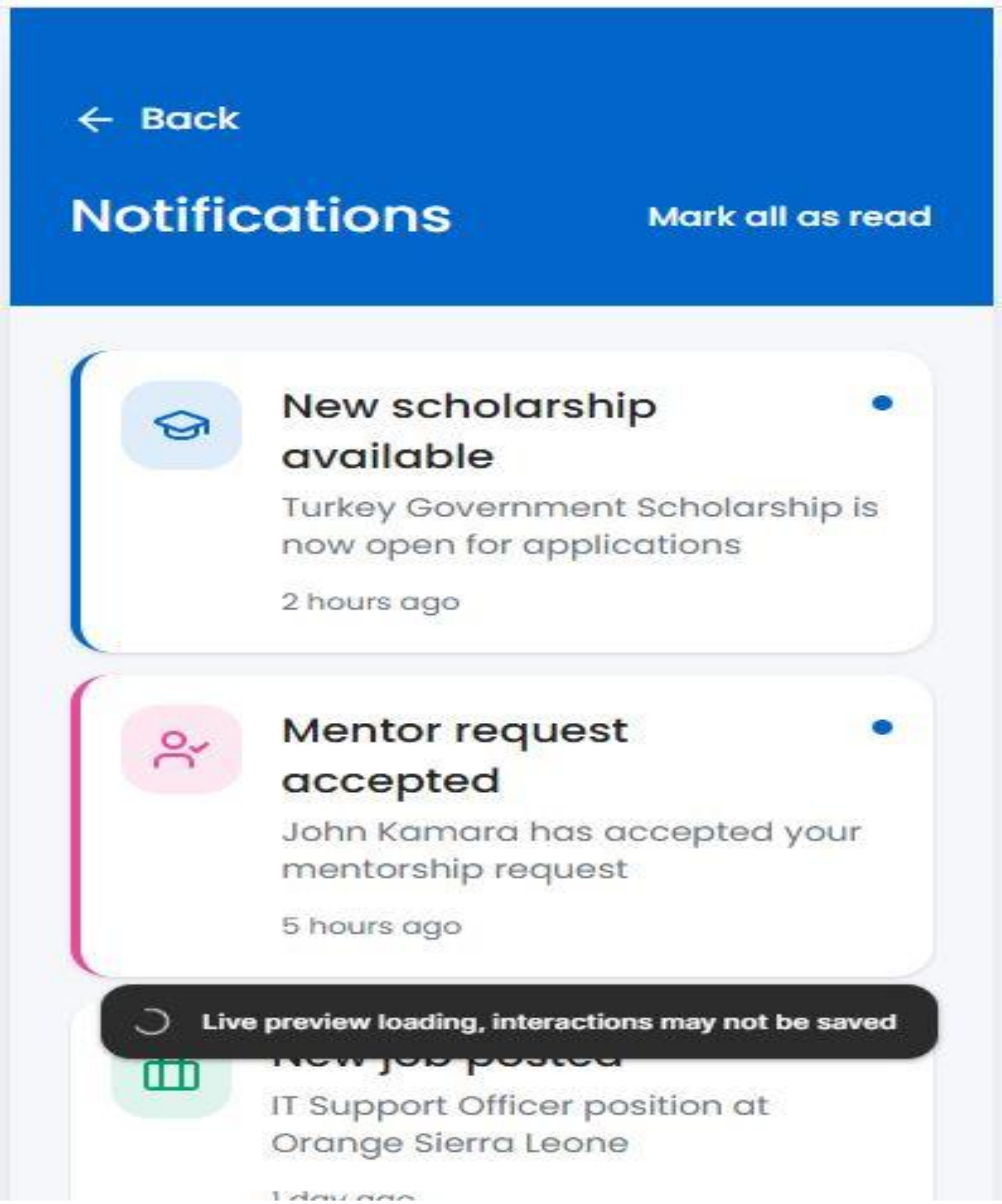

Figure 5.13: Notifications Screen

The Notifications Screen keeps users informed about important updates including new opportunities, application outcomes, and mentorship activities.

5.17 **Profile Settings Screen**

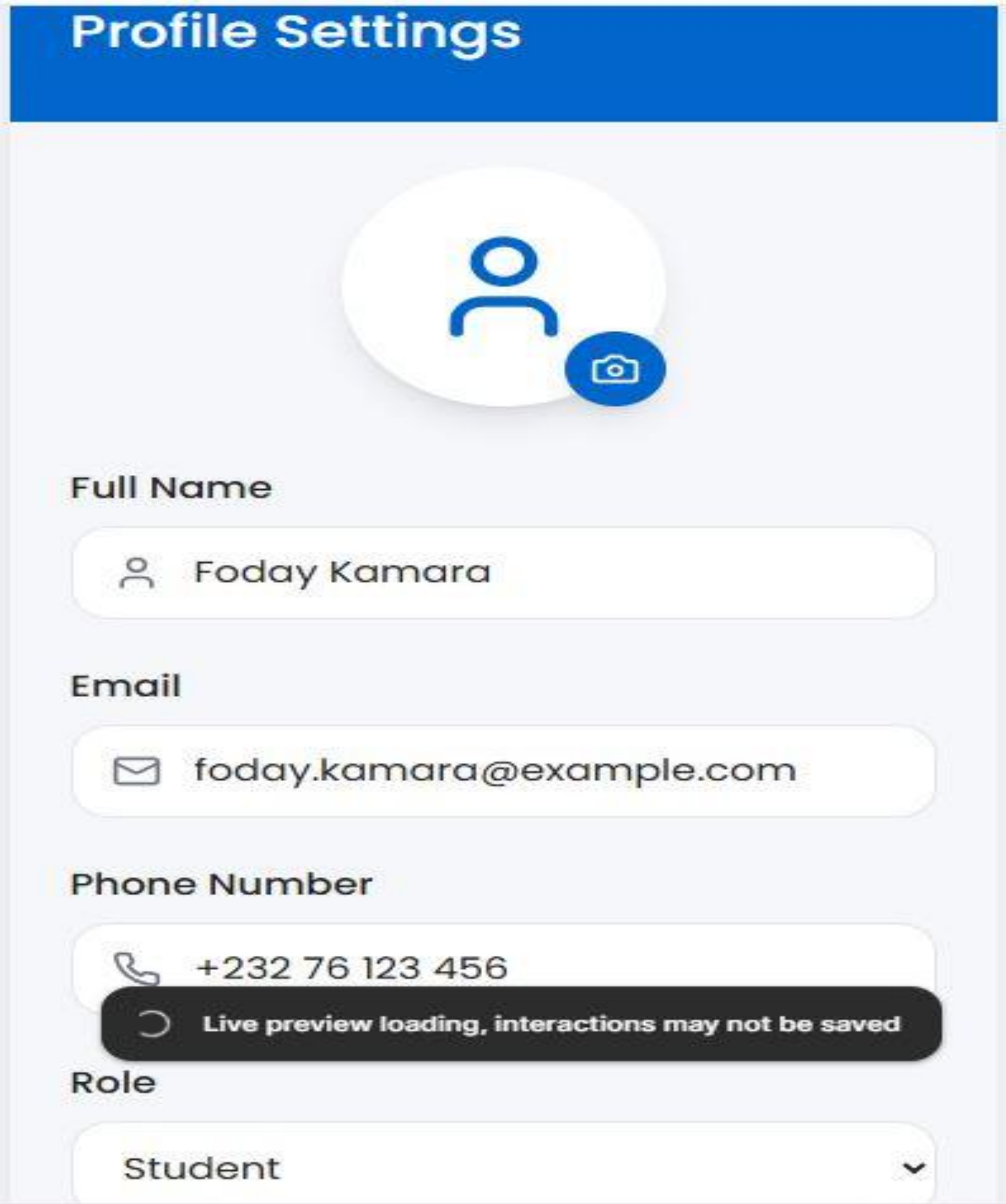

Figure 5.14: Profile Settings Screen

The Profile Settings Screen allows users to update their personal information, manage account settings, and modify security preferences.

5.18 Admin Dashboard Screen

Figure 5.15: Admin Dashboard Screen

The Admin Dashboard provides administrative functions for managing users, opportunities, educational resources, and system reports.

This interface ensures efficient platform administration and monitoring.

5.19 Prototype Navigation Flow

The prototype follows a structured navigation flow that ensures smooth movement between screens.

Users begin at the Splash Screen and proceed to the Welcome Screen before accessing Login or Registration functions.

After authentication, users are redirected to the Dashboard where they can navigate to scholarships, jobs, internships, mentorship services, learning resources, notifications, and profile management sections.

5.20 Chapter Summary

The prototype demonstrates how users interact with the platform and provides a visual representation of the proposed solution. Through modern UI/UX principles and responsive mobile design, the prototype successfully supports the objectives of the project.

CHAPTER 6: LICENSE, DESIGN QUALITY AND SDG RELEVANCE

6.1 Introduction

This chapter discusses the software license selected for the project, evaluates the quality of the system design, and explains how the proposed solution contributes to the United Nations Sustainable Development Goals (SDGs).

6.2 Software License

Selected License: MIT License

The MIT License was selected for EduConnect Sierra Leone because it is one of the most flexible and widely used open-source software licenses.

The license permits users to use, modify, distribute, and improve the software while maintaining appropriate attribution to the original creators.

The MIT License encourages innovation, collaboration, and future development, making it suitable for educational and public-good software projects.

6.3 Design Quality Evaluation

Usability

The system was designed with a user-centered approach to ensure simplicity and ease of use.

Users can easily navigate between sections using clear menus, icons, and navigation components.

Consistency

A consistent color palette, typography, layout structure, and navigation pattern were used throughout the application.

This consistency improves user familiarity and reduces learning time.

Accessibility

The interface uses readable font sizes, sufficient spacing, and clear visual hierarchy to improve accessibility for diverse users.

Scalability

The proposed architecture supports future expansion and integration of additional services, institutions, and opportunities.

Maintainability

The layered architecture and modular design approach simplify future maintenance and upgrades.

6.4 Sustainable Development Goal (SDG) Relevance

SDG 4: Quality Education

EduConnect Sierra Leone contributes to SDG 4 by improving access to educational resources, scholarships, mentorship programs, and professional learning opportunities.

The platform promotes lifelong learning and supports educational inclusion.

SDG 8: Decent Work and Economic Growth

The platform supports SDG 8 by helping students and graduates discover internships, jobs, and career development opportunities.

Improved access to employment contributes to economic growth and workforce development.

SDG 9: Industry, Innovation and Infrastructure

EduConnect Sierra Leone leverages digital technology to solve social challenges through innovation.

The platform demonstrates how software solutions can improve access to information and opportunities while strengthening digital infrastructure.

6.5 Benefits of the Proposed System

The proposed system provides numerous benefits including:

- Improved access to scholarships and educational opportunities
- Increased visibility of jobs and internships
- Enhanced mentorship and professional networking
- Centralized access to learning resources
- Improved communication between stakeholders
- Promotion of digital transformation initiatives

6.6 Chapter Summary

This chapter presented the selected software license, evaluated the quality of the proposed design, and demonstrated the project's contribution to global sustainable development goals.

The findings indicate that EduConnect Sierra Leone aligns with educational, economic, and technological development objectives while promoting digital inclusion and innovation.

CHAPTER 7: PROJECT MANAGEMENT

7.1 Introduction

Project management is an important aspect of software engineering that ensures software projects are completed successfully within the allocated time and resources.

This chapter presents the project schedule and organizational structure used during the design and development of EduConnect Sierra Leone.

7.2 **Project Schedule**

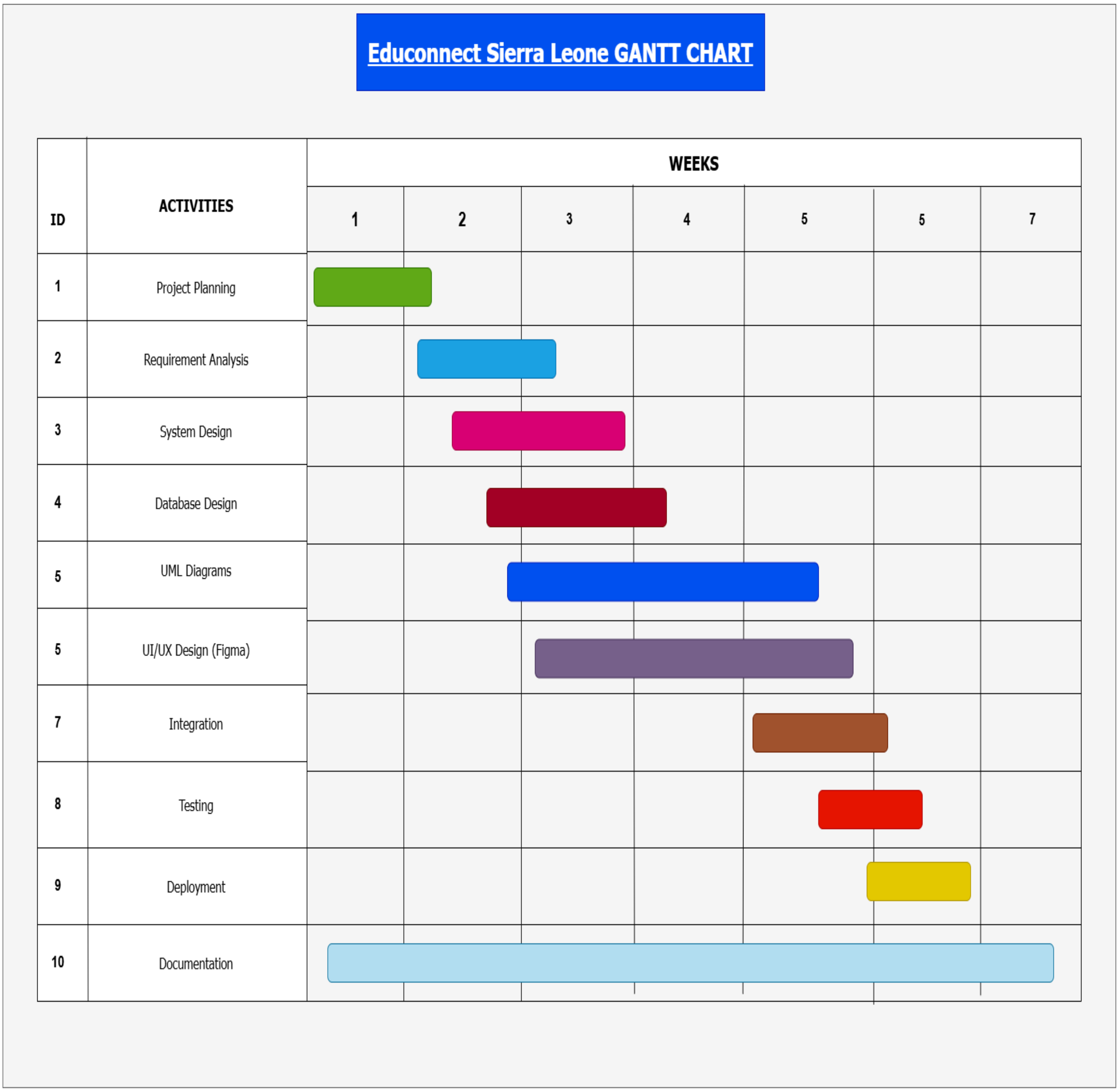

Figure 7.1: Project Gantt Chart

The Gantt Chart illustrates the timeline and sequence of activities carried out during the project.

The project was divided into several phases including planning, requirements analysis, system design, prototype development, testing, documentation, and final submission.

The schedule helped ensure that project activities were completed systematically and within the allocated academic timeframe.

7.3 **Project Activities**

Planning

The planning stage involved identifying the problem, defining objectives, determining project scope, and selecting the development methodology.

Requirements Analysis

This phase focused on identifying stakeholder needs and documenting functional and non-functional requirements.

System Design

System design involved creating UML diagrams, Data Flow Diagrams, Entity Relationship Diagrams, System Architecture Diagrams, and Network Diagrams.

Prototype Development

The prototype development phase involved designing a high-fidelity mobile application prototype using Figma.

The prototype demonstrated how users interact with the system and provided a visual representation of the proposed solution.

Testing and Review

The prototype and design models were reviewed to ensure consistency, correctness, usability, and alignment with project objectives.

Documentation

Documentation involved compiling all project activities, diagrams, explanations, and prototype screenshots into a comprehensive report.

7.4 Repository Structure

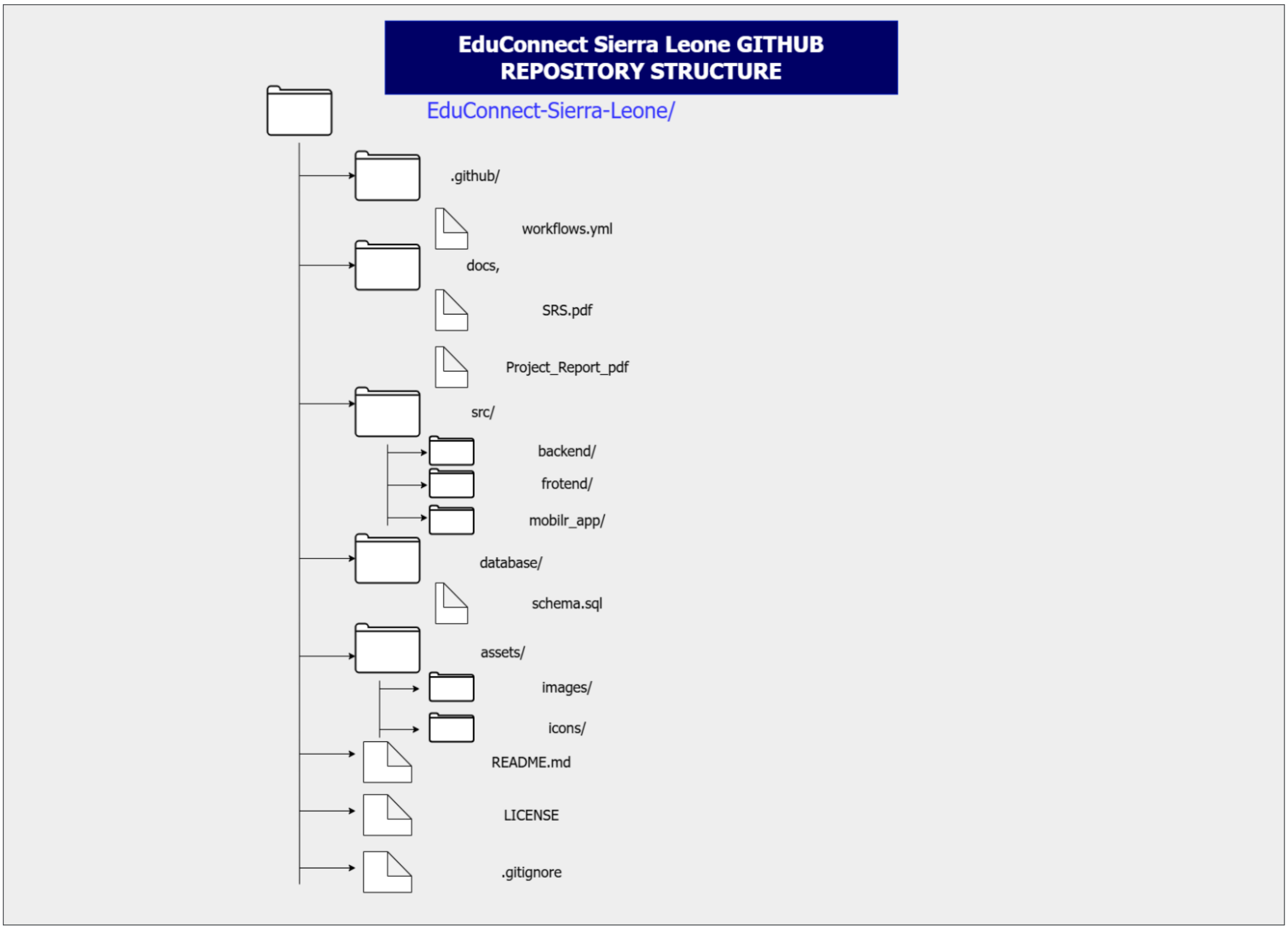

Figure 7.2: GitHub Repository Structure

The repository structure provides a logical organization of project resources and documentation.

Folders are organized according to their purpose, making the project easier to maintain, manage, and understand.

Repository Explanation

The repository contains folders for documentation, prototype files, database design artifacts, frontend resources, backend resources, and supporting assets.

This structure promotes maintainability, collaboration, and future development.

7.5 **Risk Management**

Several potential risks were identified during the project.

1. **Time Constraints**

Academic deadlines could affect project completion.

Mitigation:

A project schedule was developed using a Gantt Chart.

2. **Design Complexity**

Complex system requirements could increase development effort.

Mitigation:

The project was divided into manageable phases.

3. **Technical Challenges**

Limited access to development resources could affect implementation.

Mitigation:

Industry-standard design tools such as Draw.io and Figma were used.

7.6 Chapter Summary

This chapter presented the project management activities undertaken during the development of EduConnect Sierra Leone.

Project planning, scheduling, documentation, and repository organization contributed significantly to the successful completion of the project.

7.7 Team Member Contributions

The successful completion of EduConnect Sierra Leone required collaboration among all team members. Responsibilities were distributed according to individual strengths and expertise.

Team Member	Responsibilities
Alhaji Foday Koroma	Project Planning, Requirements Analysis, System Architecture Design, Documentation, UML Diagram Development, Repository Management
Emmanuel S Kainyande	Research, System Modeling, Testing Support, Quality Review
Robert Isaac Kamara	Prototype Design, User Interface Development, Documentation Assistance

The collaborative effort of all team members contributed significantly to the successful design and documentation of the proposed Digital Public Good platform.

CHAPTER 8: CONCLUSION AND RECOMMENDATIONS

8.1 Conclusion

This project successfully designed a Digital Public Good platform known as EduConnect Sierra Leone.

The platform was developed to address challenges faced by students and graduates in accessing scholarships, internships, employment opportunities, mentorship programs, and learning resources.

A comprehensive software engineering approach was applied throughout the project, including requirements analysis, software development life cycle modeling, system design, architecture design, network design, database modeling, and user interface prototyping.

The resulting solution provides a centralized platform that promotes educational accessibility, career development, mentorship, and professional networking.

Furthermore, the project demonstrates how technology can be used to support national development goals while contributing to Sustainable Development Goals (SDGs) related to education, employment, and innovation.

The successful completion of this project confirms the importance of software engineering methodologies in solving real-world social and educational challenges.

8.2 Recommendations

The following recommendations are proposed for future development of EduConnect Sierra Leone:

1. **Full System Implementation**

The prototype should be developed into a fully functional web and mobile application.

2. **Scholarship Integration**

The system should integrate with local and international scholarship providers.

3. **University Partnerships**

Partnerships should be established with universities and educational institutions.

4. **Employer Partnerships**

Organizations and employers should be encouraged to advertise opportunities through the platform.

5**. Artificial Intelligence Features**

Future versions may include AI-powered career guidance and recommendation systems.

6. **Mobile Application Deployment**

The platform should be deployed on Android and iOS devices.

7. **Cloud Infrastructure**

Cloud-based services should be used to improve scalability, reliability, and accessibility.

8. **Expansion Beyond Sierra Leone**

The platform may be expanded to serve neighboring countries and regional educational communities.

8.3 Future Work

Future research and development may focus on advanced features such as:

• Online mentorship sessions

• Video conferencing integration

• AI career advisors

• Automated scholarship matching

• Skills assessment tools

• Professional networking communities

• Online certification programs

These enhancements would further improve the effectiveness and impact of the platform.

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APPENDICES

Appendix A – Figma Prototype Link

The complete high-fidelity prototype for EduConnect Sierra Leone was developed using Figma.

Figma Prototype Link:

EduConnect Sierra Leone Prototype – Figma

The prototype demonstrates the complete user navigation flow and includes all major system screens designed during the project.

Appendix B – GitHub Repository Link

The project repository was created using GitHub to support version control, collaboration, documentation management, and project organization.

GitHub Repository Link:

https://github.com/alhajifodaykoroma-prog/EduConnect_SierraLeone_Development

Repository Contents:

- Project Documentation
- UML Diagrams
- DFD Diagrams
- ERD
- System Architecture Diagram
- Network Diagram
- Gantt Chart
- Figma Prototype Resources
- README File